# Simulating predictive-biomarker interactions in aggregated N-of-1 trials: an extended data-generating process building on Hendrickson et al.

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October 29, 2026

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# Abstract

**Background.** Validating a predictive biomarker requires an interaction test, which needs about four times the sample of a main-effect test (Brookes et al. 2004). Aggregated N-of-1 designs are proposed to afford it, and their power must be estimated by simulation. Hendrickson et al. (2020) proposed a simulation method for this problem. The baseline biomarker and three latent response components at every visit are drawn from one multivariate normal distribution assembled from each participant's exposure state. The predictive biomarker is a correlation with the drug response that is present on drug and absent off drug. We extend that data-generating process (DGP) for the Hybrid design with carryover and judge each extension by five criteria.

**Methods.** We derive the closed-form moments and power of a paired-difference interaction statistic under the original DGP and the extensions. We derive a Schur-complement bound on the feasible interaction strength. We verify them by Monte Carlo simulation, including runs of the original code, and test fourteen analyses against a prognostic biomarker.

**Results.** The original DGP holds the biomarker-response correlation constant after first exposure. With any carryover the interaction is then absent and power equals the test size. We extended the DGP so the correlation decays at the carryover half-life. Power then returns to its no-carryover value at half-lives of 0.1 to 0.2 week and declines monotonically at longer ones. In the original

code it raised power at 0.1 week from 0.13 to 0.64, against 0.67 without carryover. Under the original compound-symmetry structure no joint distribution exists for correlations above 0.256, so the original effect size of 0.6 relied on repaired matrices. A separable AR(1) structure at serial correlation 0.7 raises the bound to 0.474. A prognostic biomarker with a pathway absent at baseline makes analyses that share the biomarker effect with baseline anticonservative.

**Conclusions.** The correlation encoding of the interaction is coherent. It needs a decaying correlation, a correlation structure with room for realistic effects, and a feasibility check that stops instead of repairing. We give the closed-form results, the code changes, the extended DGP and planning sample sizes.

**Trial registration.** Not applicable.

**Keywords:** N-of-1 trials; predictive biomarker; simulation; carryover; data-generating process; power; prazosin.

# Background

Validating a predictive biomarker requires a biomarker-by-treatment interaction test. Such a test needs roughly four times the sample of a main-effect test of the same size (1,2). Aggregated N-of-1 designs are proposed as the way to afford it. Their power must be estimated by simulation. Hendrickson et al. (3) proposed a simulation method for this problem. To our knowledge it has no precedent in the N-of-1 literature. The baseline biomarker and, at every visit, three latent response components are drawn from one multivariate normal distribution. The trial design assembles the correlation matrix from each participant's exposure state. The predictive biomarker is a correlation with the drug response that is present on drug and absent off drug. We call this process the original DGP. This paper extends it for the Hybrid design, which combines open-label titration, blinded discontinuation and a brief crossover, with carryover between phases. The extensions address two limits of the original DGP. Under any carryover the original DGP has no interaction to detect. Its largest effect size has no valid joint distribution. The paper adds a correlation that decays after discontinuation and a feasibility check. It decides the elements the original DGP left open. It ends with the extended DGP and the sample sizes it implies.

## Aggregated N-of-1 trials and predictive biomarkers

In an aggregated N-of-1 trial each participant moves between drug and placebo, or between drug and no drug, several times. The individual series are analyzed together (4–6). Every participant contributes both an on-drug and an off-drug response. The design can therefore estimate how the drug effect varies between people. It can then test whether a baseline measurement predicts who benefits (7,8). The motivating example is prazosin for post-traumatic stress disorder (PTSD). Higher pretreatment standing systolic blood pressure was associated with greater symptom reduction on prazosin but not on placebo (9). Blood pressure is therefore a candidate predictive biomarker (10–12).

Closed-form power calculations exist only for simplified versions of these designs (13,14). Power is therefore estimated by simulating trials from a data-generating process (DGP) and analyzing each simulated trial (15). Several simulators of N-of-1 and aggregated N-of-1 data exist. Percha et al. (16) simulate single series with exponential wash-in and wash-out to choose block lengths and sampling rates. Chen and Chen (17) generate cycles under compound symmetry or AR(1) with a carryover effect to compare analyses. Blackston et al. (18) and Gärtner et al. (19) simulate aggregated series with carryover to compare designs and estimators. None of them represents a baseline biomarker that modifies the drug response. The DGP determines what a simulation can show: how the biomarker interaction is represented, how the drug effect builds and wanes, and how carryover from one period reaches the next.

## The original DGP

The original DGP has three parts. The first is a component decomposition of the symptom score. The score is the baseline level minus three responses. Each response follows a modified Gompertz curve. The drug-response component (BR) is driven by time since the current on-drug period began. The placebo-response component (PB) is driven by time in the trial and scaled by the participant's expectancy, the probability of receiving drug given the design. The natural-history component (TV) is driven by time in the trial. The second part is the design specification for open-label, crossover, blinded-discontinuation and Hybrid designs. It tracks time on drug, time since discontinuation and expectancy at each visit. The third part is open software (the `pmsimstats` R package).

The original DGP also encodes the predictive biomarker in a distinctive way. Simulation studies of

N-of-1 and crossover designs usually represent treatment-effect heterogeneity as random participant-by-treatment variation (7,20–22). The original DGP instead places the biomarker in the same multivariate normal distribution as the response components. The biomarker has a nonzero correlation $c_{bm}$ with the drug response at on-drug visits. Without carryover it has zero correlation elsewhere. We call $c_{bm}$ the biomarker-response correlation. A predictive biomarker is then one whose association with the outcome differs between on-drug and off-drug visits. The parameterization keeps the marginal distribution of the drug response fixed. It expresses the strength of the interaction as a correlation. We call this the correlation encoding of the interaction. The priority claim above rests on a PubMed search on 2026-10-27. The search was for N-of-1, single-patient or aggregated N-of-1 trials with simulation, a biomarker or effect modifier, and power or sample size. It returned one record, the original study.

## Why the joint distribution is built from the design

The simplest DGP that could serve would generate one symptom score per visit. Its mean would carry the design effects and an interaction term $\beta\, B_i D_{it}$. Its residual would be compound symmetric or AR(1). Most simulators of crossover and N-of-1 data use that form (13,17,21), and the analysis model assumes it. It is adequate when the question is the main effect. It is not adequate for the questions this series asks, for four reasons.

1. **Carryover would be whatever the analyst specifies.** In the simple form carryover is a mean shift added at off-drug visits. Whether the design still identifies the interaction under carryover depends on how the biomarker's association with the response persists after discontinuation. The simple form has no such association, so the question cannot be studied.
2. **The interaction would be only a conditional mean effect.** With a mean term the residual variance of the drug response is the same at every biomarker value. A predictive biomarker that explains part of the between-person variation in drug response implies a responder class in the population. That implies a correlation between the biomarker and the drug response. The simple form has no parameter for it (Appendix A.6).
3. **A prognostic pathway could not be generated.** A biomarker may also be associated with the placebo response or with the natural history of the condition. Such a biomarker enters the observed outcome through a pathway the analysis cannot see. Whether an analysis

keeps its nominal size under that feature can only be tested by a DGP that generates the pathways separately.

4. **Existence and identifiability would be invisible.** In the simple form every parameter combination yields a valid distribution. The design enters only through the regressor pattern. Whether the simulated effect sizes are jointly possible, and how much of the interaction the design can see, are questions about the joint distribution of the biomarker and the response components. They have closed-form answers only when that distribution is specified.

The original DGP specifies it. The biomarker and the three response components at every visit form one multivariate normal vector. The design assembles its correlation matrix. The biomarker is correlated with the drug response at on-drug visits, and at a value set by the exposure state at off-drug visits. The components are serially correlated in calendar weeks. The components are cross-correlated through a common block. Three results follow. The largest feasible interaction is a property of the design, not a number found by trial and error. We call it the feasibility bound. The power of a paired-difference statistic has a closed form. The design enters it through the on/off correlation difference, the on-drug correlation minus the mean off-drug correlation. The residual structure the analysis model should assume is derived from the DGP rather than asserted. Three requirements come with this form. The marginal covariance of the outcome is implied rather than chosen. The joint distribution must exist. Every entry must be calibrated or declared uncalibrated. The Discussion shows that, conditional on the biomarker, the correlation encoding reduces to the mean-term form with a smaller residual variance. Nothing standard is lost.

The DGP is not new in its parts. Simulating repeated measures from a multivariate normal with compound symmetry or AR(1) and a carryover term is routine (13,17). Decomposing a clinical trajectory into disease progression, placebo response and drug action is the pharmacometric convention (23). Separable (Kronecker) covariances for several response variables measured repeatedly are an established model (24). The joint correlation structure of an effect modifier and an outcome has been used to design heterogeneity analyses in cluster randomized trials (25). The original DGP combined these parts. It placed a baseline variable in the joint distribution of the latent response components, with a correlation indexed by each visit's exposure state. This paper extends that combination. We found no other instance of it in the searches reported above.

## What a data-generating process for these trials must satisfy

Reusing the original DGP for a series of simulation studies showed where it could be extended. To judge an extension, a criterion for "better" is needed. We use five. Each stage of the Results ends with the choice made and the criterion it rests on.

1. **Existence.** The joint distribution must exist for the effect sizes a study simulates. The correlation matrix must be positive definite. We measure this by the largest feasible $c_{bm}$ (the feasibility bound) and by the number of matrices a grid would need to repair.
2. **Identifiability under carryover.** The biomarker-by-treatment interaction must remain a contrast between on-drug and off-drug visits when carryover is present. We measure this by the on/off correlation difference. The closed form shows that this difference governs power.
3. **Fidelity to the pharmacology.** The time course of the drug effect, its decay after discontinuation and its behavior at re-exposure should be consistent with a short-half-life, titrated drug such as prazosin. This is a judgment. We state it as such where it decides a stage.
4. **Analyzability.** The residual structure the DGP generates should be one the analysis model can represent. The DGP should also be able to generate the features that decide which analyses are valid, such as a prognostic biomarker. We measure this by the agreement of the generated residual structure with the analysis model's, and by null rejection rates under those features.
5. **Calibration status.** Every parameter should be labeled as the original value, a choice made here, or uncalibrated. No parameter of the DGP has yet been estimated from prazosin trial data. This criterion is therefore reported, not met.

A DGP meeting criteria 1 to 4, with criterion 5 reported, is what we call the extended DGP. Its label in the repository's code and data is `refined`.

**The reference drug.** Prazosin's plasma half-life is about 2.3 hours (26), against weekly visits. The plasma concentration therefore falls by orders of magnitude within a day of the last dose. We assume a symptom-level half-life of 0.1 to 0.2 week (17 to 34 hours), or 7 to 15 plasma half-lives, as the pharmacological range. This is an assumption, not a measurement. No symptom-response half-life for prazosin in PTSD has been reported. The Results therefore cover 0 to 2 weeks and label half-lives of 0.5 week and above as behavioral scenarios. The drug is re-titrated after an interruption, which bears on how the DGP treats re-exposure (Stage 3).

### Aims

This paper extends the original DGP and judges each extension against the five criteria. The Hybrid design is the primary case. The open-label, blinded-discontinuation and crossover designs are comparators. The paper shows in closed form why power equals the test size under the original DGP when carryover is present, and why its joint distribution often does not exist. It gives the changes to the original code that add a decaying correlation and a feasibility check. It chooses, with reasons, the elements the original DGP leaves open. It collects the result as a reference specification with the stage that justifies each element. It gives the sample sizes a trialist needs to plan an aggregated N-of-1 trial under it.

## Methods

The Methods describe the original DGP and the tools used to judge each extension: the estimands, a statistic with a closed-form distribution, the feasibility bound and the simulation studies. Every choice about the DGP is made in the Results.

### The original data-generating process

We use the original DGP's notation, with a few additions. Participant $i$ is observed at a baseline visit $t = 0$ and at post-baseline visits $t = 1, \dots, T$, at calendar weeks $w_t$ (4, 8, 9, 10, 11, 12, 16 and 20 in the Hybrid design). The symptom score is

$$Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it}),$$

so a larger response lowers the score. Each component has a mean given by the modified Gompertz curve

$$G(x;\, m, d, r) = m\,\frac{\exp(-de^{-rx}) - e^{-d}}{1 - e^{-d}},$$

which the code obtains by subtracting the offset $me^{-d}$ from $m\exp(-de^{-rx})$ and rescaling, so that $G(0) = 0$ and $G(\infty) = m$. It is evaluated at a component-specific clock:

- The drug-response component (BR) at time on drug $t_{od}$, the time since the current on-drug period began, with $m = 10.99$, $d = 5$, $r = 0.42$ and SD 8. The curve reaches its inflection at 3.8 weeks, half its maximum at 4.7 weeks and 90% at 9.2 weeks.
- The placebo-response component (PB) at time in the trial, scaled by the expectancy $\eta_t$ (1 in the open-label phase, 0.5 when blinded), with $m = 6.51$, $d = 5$, $r = 0.35$ and SD $10\eta_t$.
- The natural-history component (TV) at time in the trial, with the same curve as PB and SD 10.

Off drug, the drug-response mean decays from its previous value with half-life $t_{1/2}$: $\mu_{BR,t} = \mu_{BR,t-1}\, 2^{-t_{sd,t}/t_{1/2}}$, where $t_{sd}$ is the time since discontinuation. We call this the original carryover rule. When the drug is resumed, time on drug restarts at zero.

The vector (biomarker, baseline, $TV_{i\cdot}$, $PB_{i\cdot}$, $BR_{i\cdot}$) is multivariate normal. Within each component the correlation between any two visits is $\rho_{cs} = 0.8$ (compound symmetry). We write $\rho_{cs}$ to keep $\rho$ for the AR(1) correlation per week. Between components the correlation is $c_1 = 0.2$ at the same visit and $c_\times = 0.1$ otherwise. The biomarker's correlation with $BR_{it}$ is $c_{bm}\, g_t$, where $g_t$ is the correlation profile, the fraction of $c_{bm}$ present at visit $t$. The original rule sets $g_t = 1$ wherever the drug-response mean is nonzero and 0 elsewhere. The correlation is therefore constant after first exposure. When the correlation matrix is not positive definite, the original code replaces it with a nearby valid matrix (`corpcor::make.positive.definite`, an approximation to the nearest correlation matrix (27)). No warning is issued.

The original analysis is the linear mixed model

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_D D_{it} + \beta_T w_t + \beta_{bm:D} B_i D_{it} + u_i + \varepsilon_{it},$$

with a random intercept $u_i$, the binary drug indicator $D_{it}$, and baseline included as a row with $D_{i0} = 0$. The interaction $\beta_{bm:D}$ is tested with Satterthwaite degrees of freedom (28). Participants are assigned to the design's paths in fixed numbers (18, 18, 17 and 17 in the Hybrid design at $N = 70$), not at random.

## Estimands

The target is the on-drug biomarker moderation of the drug response. This is the change in the drug effect per unit of biomarker while the drug is fully present. Without carryover it equals $\beta_{bm:D} = -c_{bm}\sigma_{BR}/\sigma_{bm}$.

Analyses that contrast on-drug with off-drug visits through a binary drug indicator estimate this quantity only when no moderation remains off drug. Under carryover they estimate an attenuated on-minus-off contrast. Analyses with an exposure-decayed indicator $D_{bc,it}$ at the true half-life recover the on-drug moderation. Table 1 states which analysis in this paper targets which quantity. The three sets of results are compared only where their targets coincide (no carryover) or where the attenuation is the object of study (Stage 2).

Table 1: The analyses used in this paper and the quantity each estimates.

| Analysis | Drug indicator | Estimates | Used in |
|---|---|---|---|
| Paired-difference statistic E9, E9b (closed form) | binary | on-minus-off contrast, $\beta_{bm:D}$ times the on/off correlation difference | Stages 1, 2, 4, 5; planning |
| The original analysis (mixed model) | binary | the same contrast | Stage 2 (original code) |
| Stress-test analyses | $D_{bc,it}$ at the true half-life | on-drug moderation $\beta_{bm:D}$ | Stage 6 |

## The paired-difference statistic

To isolate how carryover acts, we use a summary statistic whose distribution can be derived exactly (Appendix A). For each participant,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it},$$

the difference between the mean on-drug and mean off-drug scores. It is regressed on $B_i$ by least squares. We call it E9 when it uses post-baseline visits only. We call it E9b when baseline is counted

as one more off-drug visit, as in the original analysis. In the Hybrid design, Monte Carlo simulation of the original mixed model shows that E9b reproduces its mean estimates to within 0.004. It reproduces its power closely (0.654 against 0.632 without carryover). In the crossover design the closed form understates the mixed model's power (0.645 against 0.708 without carryover, and 0.279 against 0.31 to 0.35 under the original rule). The mixed model's time term uses the between-path contrast that the paired difference discards. The derivation gives three results.

- **Proposition 1.** The variance of $\Delta$ does not depend on the carryover half-life. The half-life enters only through the response means and the correlation profile. Neither appears in the variance.
- **Proposition 2.** The population slope is

$$\beta^{\mathrm{E9}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\bar{g}_{\mathrm{off}}\right), \qquad \beta^{\mathrm{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\left(1-\frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\,\bar{g}_{\mathrm{off}}\right),$$

  where $\bar{g}_{\mathrm{off}}$ is the mean of $g_t$ over the $n_{\mathrm{off}}$ post-baseline off-drug visits. The slopes are specific to each path. In the Hybrid design $n_{\mathrm{off}}$ is 3 in two paths and 4 in the other two. The pooled slope is their allocation-weighted average.
- **Power** is a noncentral-$t$ probability governed by the correlation $r_{\Delta B}$ between $\Delta$ and $B$. For E9 that correlation is proportional to the on/off correlation difference $\mathcal{G} = 1-\bar{g}_{\mathrm{off}}$. For E9b it is proportional to $1-\frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\bar{g}_{\mathrm{off}}$.

So carryover can affect power only through the on/off correlation difference. The propositions reduce the question of why power falls to the question of what the correlation rule does to $\bar{g}_{\mathrm{off}}$.

## The feasibility bound

Write the response block of the correlation matrix as $M$ and the biomarker's correlations with the response components as $c_{bm}\tilde{g}$, with $\tilde{g}$ the correlation profile padded with zeros at the $TV$ and $PB$ entries. The full matrix is positive definite if and only if $M$ is, and

$$c_{bm}^2\,\tilde{g}^\top M^{-1}\tilde{g} < 1.$$

The left side is the squared multiple correlation of the biomarker on the whole response trajectory. The largest feasible $c_{bm}$, the feasibility bound, is therefore $c^*_{bm} = (\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$ (Appendix A.4).

It depends on the correlation structure, the design and the correlation profile. It is a bound on the DGP, not on any analysis.

## Simulation studies

We designed and report the simulations following the ADEMP structure of Morris et al. (15): aims, data-generating mechanisms, estimands, methods and performance measures. There are three studies (Table 2). The scripts are listed under Availability of data and materials. The closed-form calculations of Stages 4 and 5 and of the planning section use the same code as study 1 with $\rho$ varied. A fourth run, described in the planning section, simulates the analysis G9 of the companion paper (29) under the extended DGP. It supplies the G9 columns of the planning tables and the G9 panel of Figure 4B (Figure 4).

Table 2: Design of the three simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for $R$ replicates.

| | 1. Closed-form verification | 2. The original code | 3. Stress test under a prognostic biomarker |
|---|---|---|---|
| **Aim** | Check the closed-form moments and power of Appendix A, and the loss of power under the original rule | Show on the original code that the correlation rule alone determines the loss of power | Find which analyses keep their nominal size under features the extended DGP adds |

| | 1. Closed-form verification | 2. The original code | 3. Stress test under a prognostic biomarker |
|---|---|---|---|
| **Data-generating mechanisms** | The correlation encoding with the original rule and with the decaying correlation; compound symmetry and the separable AR(1) form; carryover on the drug-response mean decaying from the last on-drug value; Hybrid, crossover and OL+BDC designs; $N = 70$; grids of $c_{bm}$ and $t_{1/2}$ | The original `generateData()` at commit 58b32a9, as published and with the decaying correlation only (Appendix B.1); $c_{bm} = 0.25$. (a) Hybrid, $N = 70$, no censoring, $t_{1/2} \in \{0, 0.1, 0.2, 1\}$; (b) four designs, $N \in \{35, 70\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$, the original censoring patterns | The extended DGP (Table 11) with fixed allocation; base cell separable AR(1), $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week; 15 null cells (11 without and 4 with a prognostic pathway of strength 0.3) and 10 alternative cells at $c_{bm} = 0.30$ varying $N$, half-life, decay form, dropout and measurement error |
| **Estimand** | $\beta_{bm:D}$, and the mean of the paired-difference statistic | $\beta_{bm:D}$ | $\beta_{bm:D}$ |
| **Methods** | Paired-difference statistics E9 and E9b, closed form and simulated; the original mixed model fitted with `lmer` | The original analysis pipeline (`lmer`, model-based test) on both versions | Fourteen analyses fitted to the same simulated trials; two-sided tests at $\alpha = 0.05$ |
| **Performance measures** | Mean, SD and power of the statistic; agreement as (simulated − closed form)/MCSE | Power; mean interaction estimate; number of repaired correlation matrices | Null rejection rate; power; $\kappa$, the ratio of mean model SE to empirical SD; failed fits |

| | | | |
|---|---|---|---|
| **Replicates** | 5,000 per cell (MCSE of power at most 0.007); 500 to 1,000 for `lmer` | (a) 500 per cell (MCSE at most 0.022); (b) 250 per cell (at most 0.032) | 1,000 per null cell (MCSE 0.0069 at a rate of 0.05), 500 per alternative cell (at most 0.022) |

**Number of replicates.** Replicate counts were set by the precision each comparison needs. In study 3, 1,000 null replicates give a two-MCSE band of 0.036 to 0.064 for a single cell. Pooling 11 non-prognostic null cells gives the band of 0.046 to 0.054 used in Stage 6. In study 1, a cell is flagged when the simulated and closed-form values differ by more than 2 MCSE. Chance alone produces this in about 4.6% of cells. The 278 cell-variants are not independent, since each uses its own draws but the same designs.

**Random numbers.** Every cell, or chunk of a cell, sets its own seed from a fixed rule: a function of the cell's parameters in study 1, a shared seed per half-life (study 2a) or per design, sample size and half-life (study 2b), and $20261007 + 1000 \times \text{cell} + \text{chunk}$ in study 3. Results therefore do not depend on the number of cores or the order in which chunks run. Both versions of the original code in study 2 use the same seeds. All analyses in study 3 are fitted to the same trials. The comparisons are therefore paired. R's default Mersenne-Twister generator was used throughout.

**Failed fits and repaired matrices.** In study 3, a fit that returned an error was recorded as missing. Failures numbered 0 to 12 per analysis over 28,000 trials, and at most 0.6% in any cell. In study 2 the original pipeline returned an estimate for every replicate. No correlation matrix was repaired in either version, because $c_{bm} = 0.25$ lies below the compound-symmetry feasibility bound of 0.256. Study 1 does not fit models for the paired-difference statistic, so it has no failures.

**Which form of the DGP each study used.** Study 1 and the closed-form calculations build the cross-component correlation as $c_1 \rho^{|w_t - w_s|}$ (the separable form) where the separable structure is named, and as the original constants where compound symmetry is named. Study 2 is the original code and so the original structure. Study 3's driver builds the separable form with fixed path allocation (18, 18, 17, 17) and the original carryover rule on the mean. The tables reported here are from that run. A rerun with carryover decaying from the last on-drug mean and randomized allocation changed no null rejection rate beyond Monte Carlo error and no prognostic verdict (Appendix B.5).

**Software.** R 4.6.1, with lme4 2.0.1 and lmerTest 3.2.1 for the original pipeline, nlme 3.1-169, mmrm 0.3.17, clubSandwich 0.7.0 and MASS 7.3-65.

**Exploratory status.** The stress test is exploratory. Its cells and analyses were written down as a working protocol in `docs/46` of the repository. No analysis plan was fixed in advance, and the protocol was revised as results came in. Two cells with a prognostic association between the biomarker and baseline severity, and three analyses, were added after the first results. Failed fits were first counted as non-rejections and are now excluded from the denominator, a change of at most 0.0025 in any rate. The study also compared the analyses' power and selected one by a decision rule. That selection was revised after results, is post hoc, and is reported in the project's documentation (`docs/56`), not here. This paper uses the stress test only for what it shows about the DGP: which analyses are valid under a prognostic biomarker, and what the optional features do to size and power.

# Results

The Results are organized as stages. Each stage states what the original DGP does, what the extension adds, the evidence, and the choice made with the criterion it rests on. Table 11, at the end, collects the choices.

## Stage 1. Validity of the joint distribution

The original DGP accepts any $c_{bm}$ and repairs a correlation matrix that is not positive definite. Under its compound-symmetry structure the matrix is positive definite only for $c_{bm}$ below a bound. Without carryover the bound is 0.256 in the Hybrid design (0.256 in the crossover and 0.262 in the open-label-plus-blinded-discontinuation design). These values follow from the feasibility bound with $M$ the original response block. The nonzero effect sizes in the original study were 0.3 and 0.6. Counting one matrix per design path, effect size and half-life, 37% of the original grid's matrices at those effect sizes (20 of 54) were replaced by nearby valid matrices (Table 3). At $c_{bm} = 0.3$ the repair changed no correlation by more than 0.007. The original results at that effect size are therefore essentially those of the stated DGP. At $c_{bm} = 0.6$ the repair lowered the on-drug correlation to 0.54 to 0.57 and moved other correlations by up to 0.10. Those results describe a different DGP from the one stated.

The bound is the condition that the biomarker's squared multiple correlation on the whole response trajectory stays below one. Under compound symmetry every visit's drug response shares the same

person-level factor. The biomarker therefore cannot be correlated with the on-drug visits alone without being correlated, through that factor, with the off-drug visits as well. The bound is low because the correlation profile switches between visits that the structure ties together.

We extended the DGP to check feasibility and stop. When the matrix is not positive definite, the extended DGP reports the cell as infeasible instead of changing it (Appendix B.2). The package function `buildSigma()` now does so by default (Appendix B.5). The choice rests on the existence criterion. One consequence is that the original effect size of 0.6 cannot be simulated under any correlation encoding in this paper. Table 3 shows it infeasible in 75% of the extended grid under the extended DGP, and in every Hybrid cell. The original DGP passes those cells at positive half-lives because its rule keeps the correlation constant after first exposure. A constant correlation places a small demand on the structure, and it carries no interaction (Stage 2). The cells that need no repair under the original rule are the cells in which the interaction is absent. What the bound means in outcome units, and what to do above it, are taken up in the planning section and the Discussion.

The case that fixed the default was a prognostic scenario in the companion study (29), not the original grid. Two runs combined $c_{bm} = 0.45$ with a prognostic correlation of 0.6. The joint bound of Stage 6 shows that combination to be infeasible at any $c_{bm}$. The repair lowered the on-drug correlation, and the runs reported the attenuated interaction as a loss of power to the prognostic pathway. A repaired matrix does not report which correlation it changed. A repair therefore turns an infeasible input into a plausible but wrong finding. One degenerate case is distinguished from a failure of existence. The placebo-response component has standard deviation proportional to the expectancy. A visit with expectancy zero has exactly zero placebo-response variance and makes the covariance matrix singular by construction. The stop path factors only the positive-variance sub-block and draws the degenerate components at their means. No correlation is changed (Appendix B.2).

## Stage 2. The biomarker-response correlation under carryover

The original study reported that power to detect the interaction fell sharply once carryover was introduced. In our 500-replicate rerun of the original code in the Hybrid design at $N = 70$ and $c_{bm} = 0.25$ (Table 5), power fell from 0.668 without carryover to 0.126 at a half-life of 0.1 week. No correlation matrix was repaired in that run, because 0.25 lies below the compound-symmetry

Table 3: Correlation matrices that are not positive definite. Figure 4 grid: open label, open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2\}$ week. Extended grid: the last three designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The original-DGP counts reproduce those of the original code exactly. The extended-DGP counts are the same at $\rho = 0.7$ (the recommended value) and $\rho = 0.8$ because both feasibility bounds lie between 0.3 and 0.6.

| Grid (number of matrices) | Original DGP (constant correlation, compound symmetry, $\rho_{cs} = 0.8$) | Extended DGP (decaying correlation, separable AR(1)), $\rho = 0.8$ | Extended DGP, $\rho = 0.7$ |
|---|---|---|---|
| Figure 4 grid of the original study, $c_{bm} \in \{0, 0.3, 0.6\}$ (81) | 20 (25%) | 21 (26%) | 21 (26%) |
| Extended grid, $c_{bm} = 0.25$ (40) | 0 (0%) | 0 (0%) | 0 (0%) |
| Extended grid, $c_{bm} = 0.3$ (40) | 12 (30%) | 0 (0%) | 0 (0%) |
| Extended grid, $c_{bm} = 0.6$ (40) | 12 (30%) | 30 (75%) | 30 (75%) |

feasibility bound of 0.256. At that half-life only 0.1% of the drug effect remains a week after stopping. Figure 4 of the original study shows the same pattern. We did not read values off it.

Under the original rule every off-drug visit in the Hybrid design follows an exposure. With any positive half-life its drug-response mean is positive, so $g_t = 1$. Hence $\bar{g}_{\text{off}} = 1$ and, by Proposition 2,

$$\beta^{\text{E9}}_{bm:D} = 0, \qquad \beta^{\text{E9b}}_{bm:D} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \frac{1}{n_{\text{off}} + 1} \qquad \text{for every } t_{1/2} > 0.$$

Power is a step function of the half-life (Table 4, Figure 1). It takes its no-carryover value at $t_{1/2} = 0$. It takes a constant value at every half-life above about 0.007 week, whether 0.1 week or 10. Below that, the off-drug mean underflows to zero in double precision and the original rule sets $g_t = 0$ at late off-drug visits (with the original carryover rule, below about 0.012 week). For E9b the constant is 0.081. This value comes from the baseline row, which alone keeps $g_t = 0$.

The original analysis behaves the same way. In our reconstruction of the original DGP, the mean estimate falls from −0.126 to between −0.026 and −0.032 at every positive half-life. Power falls from 0.63 to between 0.08 and 0.11. The separately seeded run of the original code reported above gives 0.126 at 0.1 week. The two implementations differ at 0.1 week by about 2.4 combined Monte

Table 4: Closed-form power by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry. Constant: the original rule (constant correlation after first exposure). Decaying: the extended rule (correlation decaying at the carryover half-life). E9: paired-difference statistic on post-baseline visits; E9b: the same with baseline counted as an off-drug visit, as in the original analysis. Against Monte Carlo estimates (5,000 replicates per cell), 10 of 278 cell-variants differ by more than 2 Monte Carlo SE.

| Correlation rule | Statistic | 0 | 0.1 | 0.2 | 0.5 | 1 | 2 |
|---|---|---|---|---|---|---|---|
| Constant (original) | E9 | 0.721 | 0.050 | 0.050 | 0.050 | 0.050 | 0.050 |
| | E9b | 0.654 | 0.081 | 0.081 | 0.081 | 0.081 | 0.081 |
| Decaying | E9 | 0.721 | 0.721 | 0.712 | 0.631 | 0.473 | 0.278 |
| | E9b | 0.654 | 0.653 | 0.647 | 0.586 | 0.472 | 0.327 |

Carlo SE, which is nominally significant. We did not trace its source. It does not bear on the conclusion, since both values are at or near the test size.

With a constant correlation, the biomarker moderates the on-drug and off-drug responses equally. No interaction remains to detect. The loss of power under small carryover is therefore a property of the correlation rule, not of carryover or of the designs.

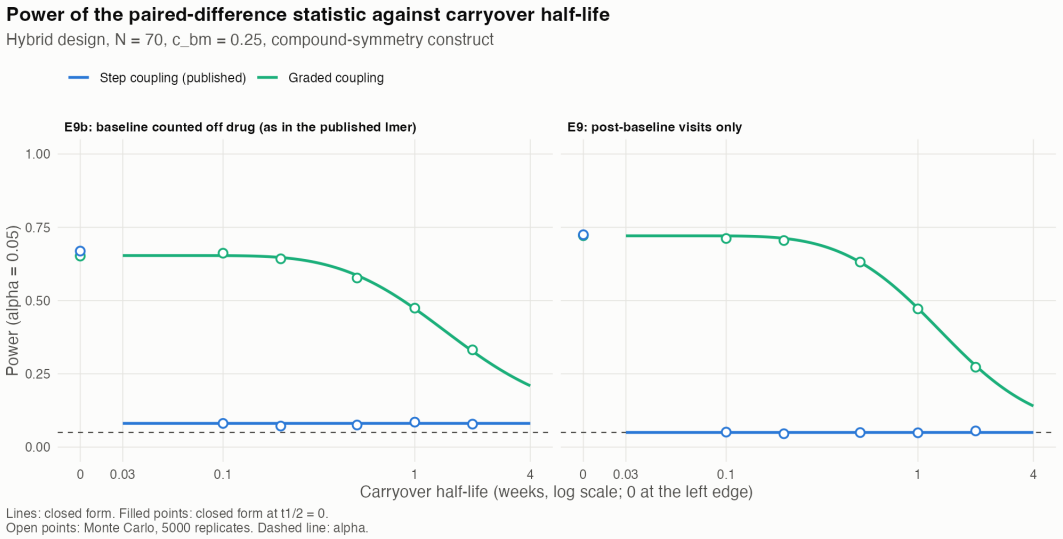



Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points; 5,000 replicates per cell), under the original rule (constant correlation after first exposure) and the extended rule (correlation decaying at the carryover half-life). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry; left panel E9b, right panel E9.

We extended the DGP so the correlation decays after discontinuation at the carryover half-life: $g_t = 1$ on drug, $2^{-t_{sd,t}/t_{1/2}}$ off drug after exposure, and 0 before any exposure. This is the pharmacological reading of a predictive biomarker. The biomarker predicts the size of the drug effect, so it cannot

predict a drug effect that is no longer present. Under the extended rule the mean of $g_t$ over the off-drug visits is

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} 2^{-t_{sd,t}/t_{1/2}},$$

which is strictly increasing in $t_{1/2}$, because every off-drug visit has $t_{sd} \geq 1$ week. Two consequences follow.

- **Power is strictly decreasing in the half-life** for the path-stratified paired-difference statistic. This holds on every path with at least one off-drug visit after an exposure (Proposition 3, Appendix A.3). The placebo-first crossover path, whose off-drug visits all precede exposure, has constant power. For the pooled statistic power is non-increasing at every step of a 120-point grid from 0.03 to 4 weeks. For the original mixed model it was checked only at the simulated half-lives.
- **The loss vanishes faster than any power of the half-life as $t_{1/2} \to 0$.** At 0.1 week, $\bar{g}_{\text{off}} \leq 2^{-10}$ and power is unchanged to within 0.001.

The original rule is the limit of the extended rule as $t_{1/2} \to \infty$, applied at every positive half-life.

Power depends on the correlation profile only through the on/off correlation difference $\mathcal{G} = 1 - \bar{g}_{\text{off}}$ (Figure 2). The difference is 1 without carryover. Under the extended rule in the Hybrid design it is 0.9997 at 0.1 week, 0.991 at 0.2, 0.906 at 0.5, 0.754 at 1 and 0.551 at 2. Under the original rule it is 0 at every positive half-life. The noncentrality is proportional to $r_{\Delta B}/\sqrt{1 - r_{\Delta B}^2}$, and $r_{\Delta B}$ is proportional to the difference. For small $r_{\Delta B}$ the sample size needed for a given power therefore scales approximately as $1/\mathcal{G}^2$. At 1 week this is about 1.8 times the no-carryover sample. At 2 weeks it is about 3.3 times.

Replacing only the correlation lines of the original code with the extended rule (Appendix B.1) and running the pipeline unchanged gave the results in Table 5. The two versions coincide without carryover, as they must. At the original half-lives the extended version keeps the full interaction, within Monte Carlo error of the no-carryover value of 0.668. At 1 week it shows a gradual decline. No matrix was repaired in either version, because $c_{bm} = 0.25$ is below the feasibility bound.

Figure 3 repeats Figure 4B of the original study on the original code, for all four designs, both sample sizes and the five censoring patterns, as published and with the decaying correlation. It uses

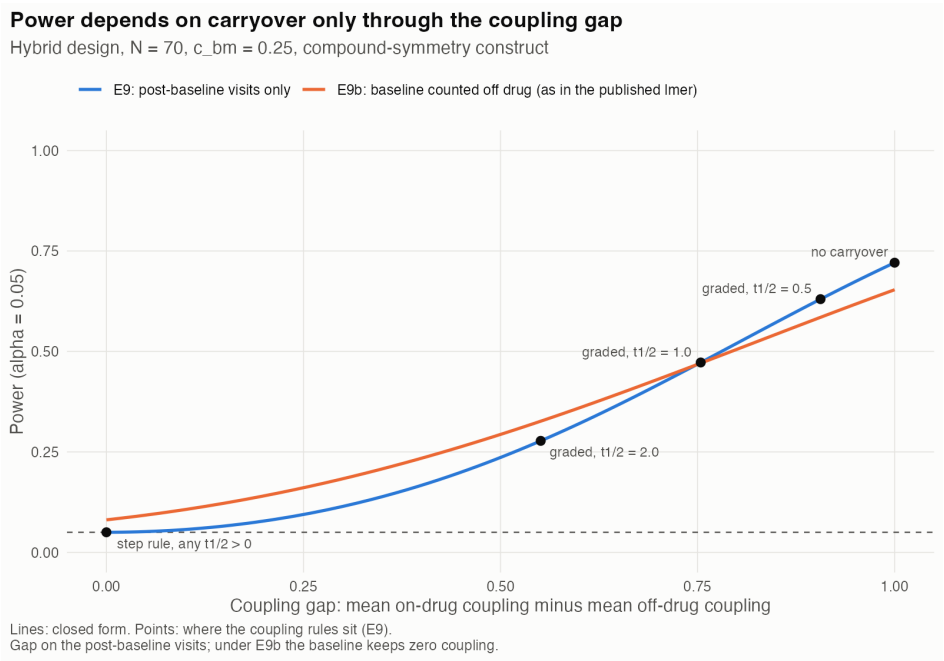



Figure 2: Closed-form power against the on/off correlation difference, with the positions of the two correlation rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$, E9.

Table 5: The original code at commit 58b32a9, as published (original) and with the decaying correlation (decaying): Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022.

| $t_{1/2}$ (weeks) | Mean estimate, original | Mean estimate, decaying | Power, original | Power, decaying |
|---|---|---|---|---|
| 0 | -0.131 | -0.131 | 0.668 | 0.668 |
| 0.1 | -0.028 | -0.129 | 0.126 | 0.638 |
| 0.2 | -0.029 | -0.131 | 0.092 | 0.656 |
| 1 | -0.033 | -0.107 | 0.094 | 0.462 |

$c_{bm} = 0.25$ instead of the original 0.6. The value 0.6 lies above every feasibility bound in this paper and would confound the comparison with the repair. At 0.25 neither version repaired a matrix, so the correlation rule is the only difference. With 250 replicates per cell the Monte Carlo SE reaches 0.032. Differences of 0.05 between adjacent cells are therefore not interpretable. The pattern is. Results at $N = 70$ without censoring:

- **Hybrid design.** Power under the original rule falls from 0.66 to 0.08 at both positive half-lives. Power under the extended rule stays at 0.62 and 0.68.
- **Open label plus blinded discontinuation.** Power under the original rule falls from 0.57 to 0.12 and 0.13. Power under the extended rule stays at 0.57 and 0.60.
- **Crossover.** Power under the original rule falls only to 0.34 and 0.33. The placebo-first path's off-drug visits precede any exposure and so keep zero correlation under both rules. Power under the extended rule stays at 0.71 and 0.78.
- **Open label.** There are no off-drug visits. The two rules coincide and power is low throughout.

The same pattern holds at $N = 35$ and under every censoring pattern.

Figure 4 repeats the grid of Figure 3 under the extended DGP of Stage 7 with the analysis G9 that the companion paper recommends (29). G9 is a random-intercept model with CAR(1) residual correlation and an exposure-decayed drug indicator whose half-life is chosen by AIC, with CR2 cluster-robust standard errors (planning section). The data-generating process differs from that of Figure 3 in its correlation structure (separable AR(1) at $\rho = 0.7$ instead of compound symmetry), so the two figures are not matched cell by cell. Each cell has 250 replicates and a Monte Carlo SE of at most 0.032. The open-label design is not estimable under G9: every visit is on drug, so the exposure-decayed indicator is constant at 1 and the interaction term is collinear with the biomarker. The original analysis replaces the interaction with a biomarker-by-time term in that case. G9 has no such substitute, and the cells are marked as not estimable rather than simulated. The censoring patterns in this figure use the corrected dropout rule. The original `censordata()` raised dropout with improvement, the reverse of the intended mechanism, through a sign error in its biased term (Appendix B.5). Under the corrected rule, dropout rises with worsening. Figure 3 keeps the published rule, because it reproduces the original code. Results at $N = 70$ without censoring:

- **Hybrid design.** Power is 0.66, 0.65 and 0.65 at half-lives of 0, 0.1 and 0.2 week.

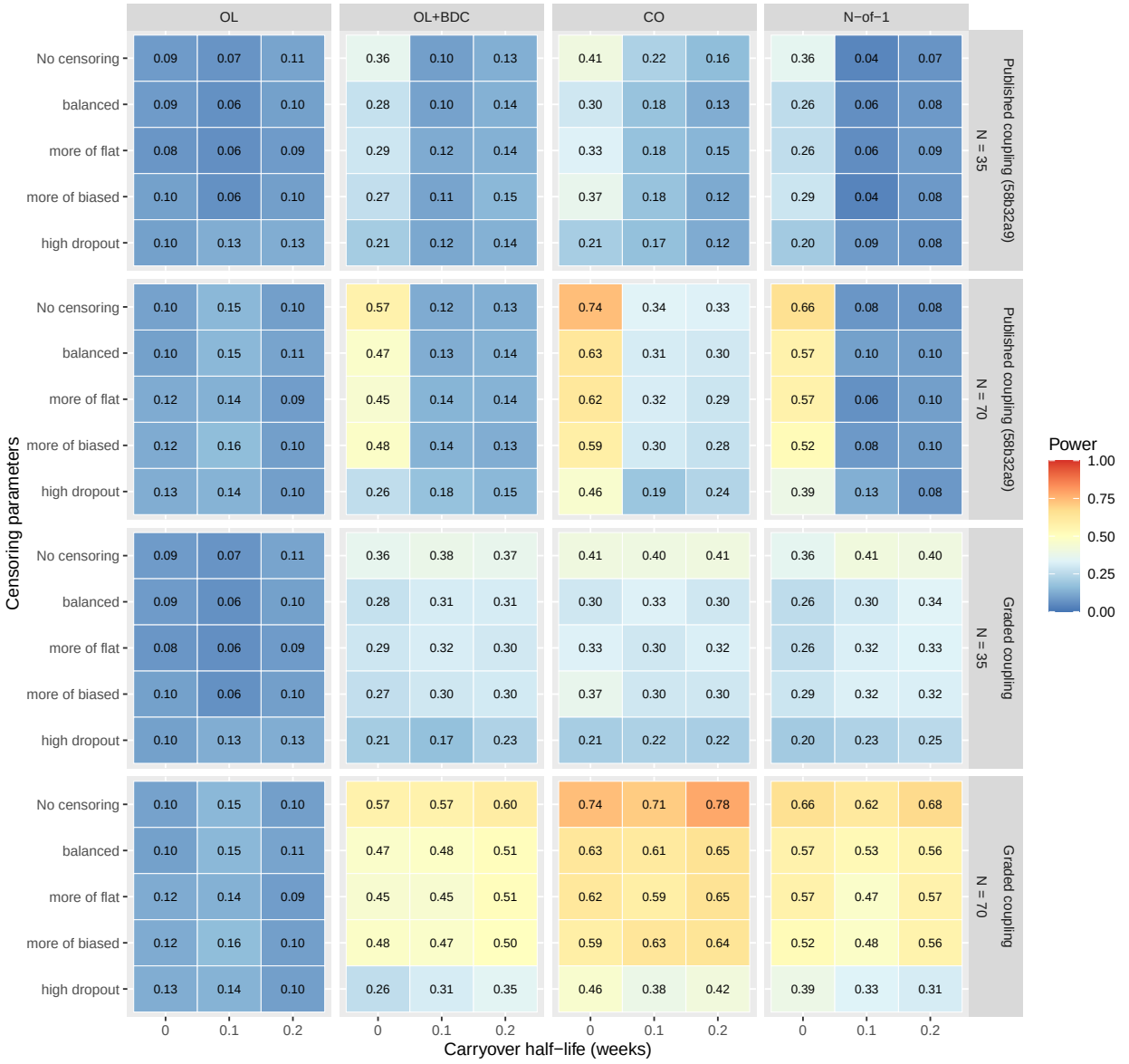



Figure 3: Power to detect the biomarker-by-treatment interaction, in the layout of Figure 4B of the original study, from the original code at commit 58b32a9 as published (upper two panel rows) and with the decaying correlation (lower two panel rows). Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions. No matrix was repaired in either version.

- **Open label plus blinded discontinuation.** Power is 0.58, 0.58 and 0.46.
- **Crossover.** Power is 0.34, 0.38 and 0.45.

Under G9 and the extended DGP, power at the pharmacological half-lives stays within Monte Carlo error of its no-carryover value in the Hybrid and OL+BDC designs. The high-dropout censoring pattern lowers power most, to 0.31 in the Hybrid design at 0.2 week. The crossover design has lower power under G9 than under the original analysis on the original code (Figure 3). The crossover design has a constant expectancy, and G9 estimates the half-life from a design with only one switch per path. We do not separate the contribution of the correlation structure from that of the analysis here.

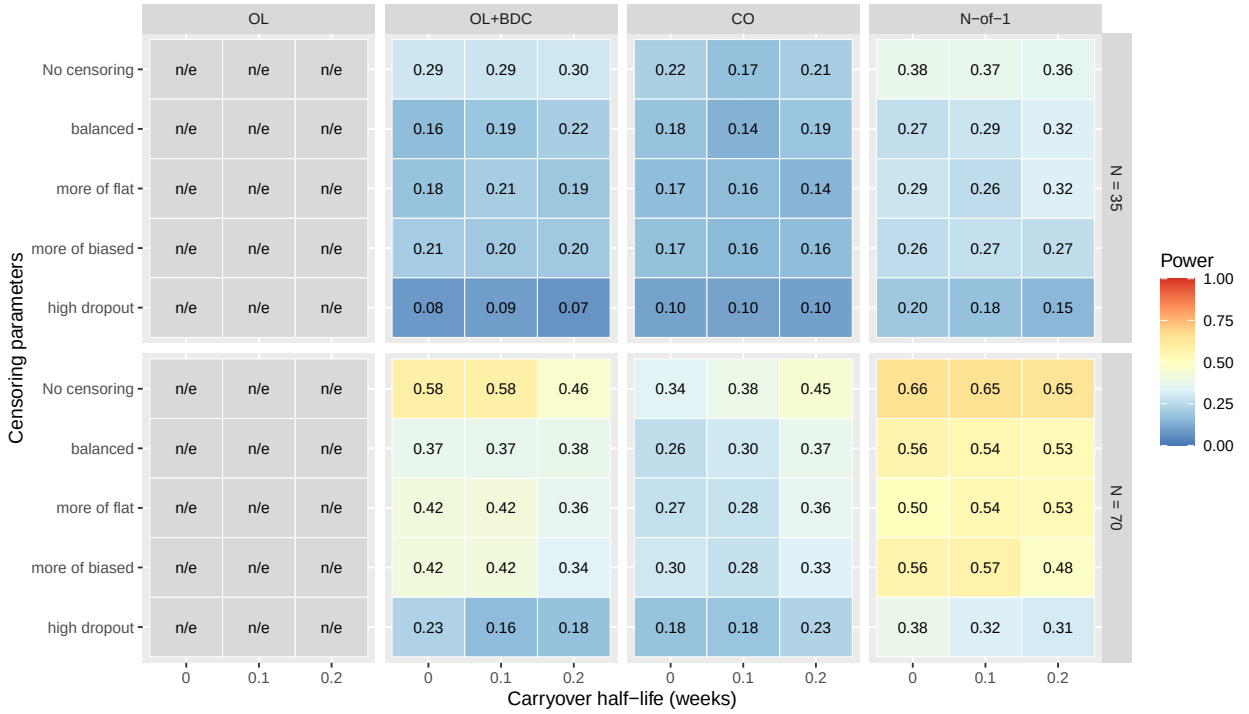



Figure 4: Power of the analysis G9 to detect the biomarker-by-treatment interaction under the extended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean), in the layout of Figure 4B of the original study. Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.032). Censoring patterns use the corrected dropout rule, under which dropout rises with worsening; the original-code figure uses the published rule of commit 58b32a9. Gray cells (n/e): the open-label design, in which every visit is on drug and the interaction is not estimable under G9.

Figure 5 extends the comparison over interaction strengths and longer half-lives. It shows closed-form power in the Hybrid design for each correlation rule.

- **The original rule.** Every column after $t_{1/2} = 0$ falls to the test size or just above it, at every strength. For example, at $c_{bm} = 0.2$ power under E9b falls from 0.46 to 0.07.
- **The extended rule.** Power declines smoothly: 0.46, 0.46, 0.46, 0.41, 0.32 and 0.23 at the same strength over half-lives of 0 to 2 weeks.
- **Feasibility.** Gray cells mark effect sizes at or above the compound-symmetry feasibility bound, where no joint distribution exists. Under the extended rule, $c_{bm} = 0.3$ becomes feasible only from $t_{1/2} = 1$ week. A correlation that decays instead of switching places a smaller demand on the structure (Appendix A.5).

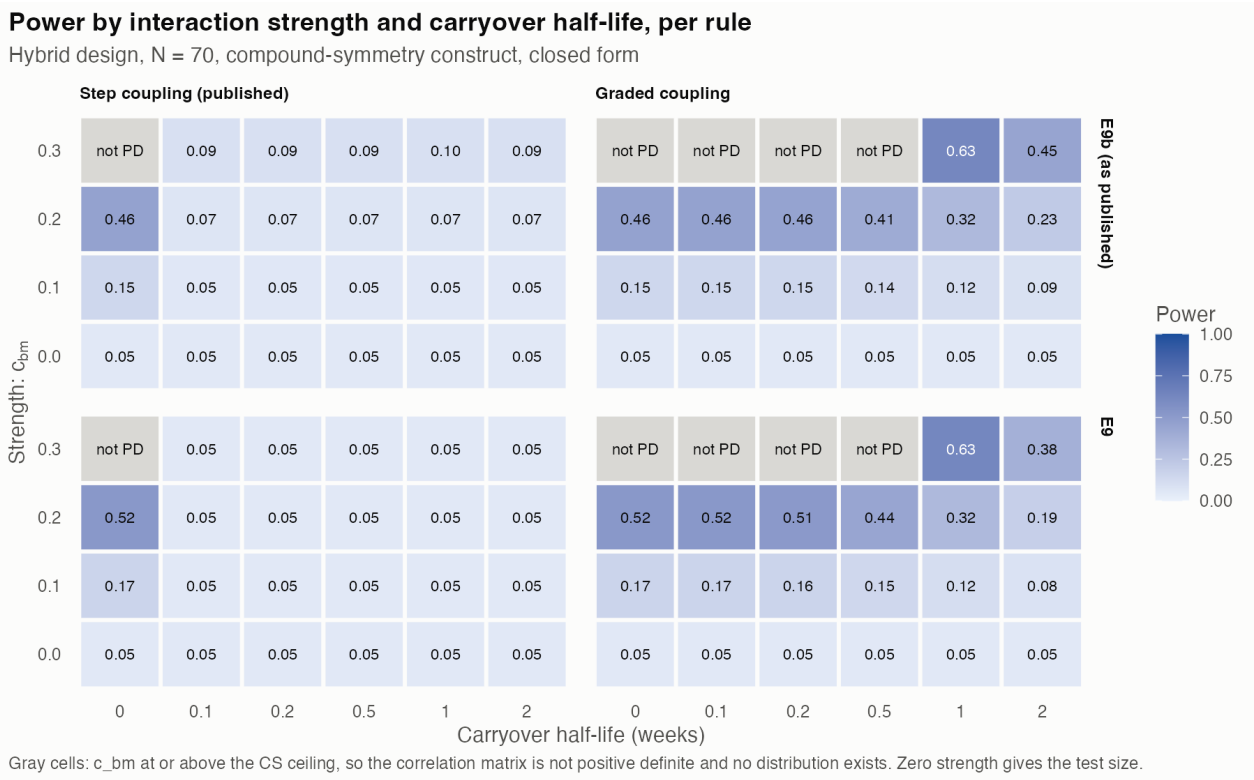



Figure 5: Closed-form power of the paired-difference statistic by interaction strength ($c_{bm}$) and carryover half-life, in the layout of Figure 4B of the original study: the original rule (labeled Step in the figure) and the decaying correlation (labeled Graded). Upper row: E9b, which counts baseline as an off-drug visit as the original analysis does; lower row: E9. Gray cells: $c_{bm}$ at or above the compound-symmetry feasibility bound, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry.

The extended DGP uses the decaying correlation. The choice rests on identifiability under carryover and on fidelity to the pharmacology. Two limitations of the choice are stated here. First, at the pharmacological half-lives of 0.1 to 0.2 week the on/off correlation difference is within 0.01 of 1. The extension therefore returns the no-carryover result. The extended DGP at its default half-life

is not a test bed for carryover. The smooth decline is a statement about half-lives of 0.5 week and above, which we label behavioral. Second, the decaying correlation is asymmetric. Off drug the moderation fades with the drug effect. On drug it is the full $c_{bm}\sigma_{BR}b_i$ at every visit, including the first weeks of titration when the Gompertz mean is a small fraction of its maximum. A correlation proportional to the current drug-response mean at every visit, or a correlation half-life separate from the mean's, would remove the asymmetry. Neither was tried. The sensitivity model of the Discussion is the form that addresses it. The mixed-model power of the extended DGP at half-lives of 0.5 to 2 weeks, with the half-life fixed in the analysis or chosen from the data, is the subject of a companion paper (29).

## Stage 3. Carryover on the mean

The original carryover rule decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. Consider two off-drug visits at weeks 11 and 12 following discontinuation at week 10. The original code decays the week-10 value by $2^{-1/t_{1/2}}$ and then by $2^{-2/t_{1/2}}$, a total exponent of 3 weeks for 2 weeks elapsed. Whether this was intended is not stated in the original study, and we do not claim that it was not. We treat the two forms as two specifications.

The extended DGP decays the mean from the last on-drug value, $\mu_{BR,\text{last}}\,2^{-t_{sd}/t_{1/2}}$. The mean at any off-drug visit then depends only on the elapsed time (Appendix B.3).

The choice changes no slope, variance or feasibility bound. It changes only the path means. It therefore changes the closed-form power of study 1 by at most 0.002 in the Hybrid design. In the original pipeline it changes power by at most 0.006. Figure 6 shows the drug-response mean by visit for the four Hybrid paths under both forms. Two features are visible. At $t_{1/2} = 1$ week the original rule lies below the extended rule at the second off-drug visit, as expected. At 0.2 week both are indistinguishable from zero one week after discontinuation. At re-exposure (week 16 in paths 1 and 3) time on drug restarts at zero. The mean then rises along a fresh Gompertz curve to 4.3 points at the week-16 visit, four weeks into the restarted curve. The residual from the earlier exposure has decayed to 0.16 points by then even at a 1-week half-life. No downward step therefore appears on the Hybrid schedule. A step would appear only if a visit fell within about a week of re-exposure at a half-life of a week or more.

The extended DGP decays carryover from the last on-drug mean. It keeps the original restart

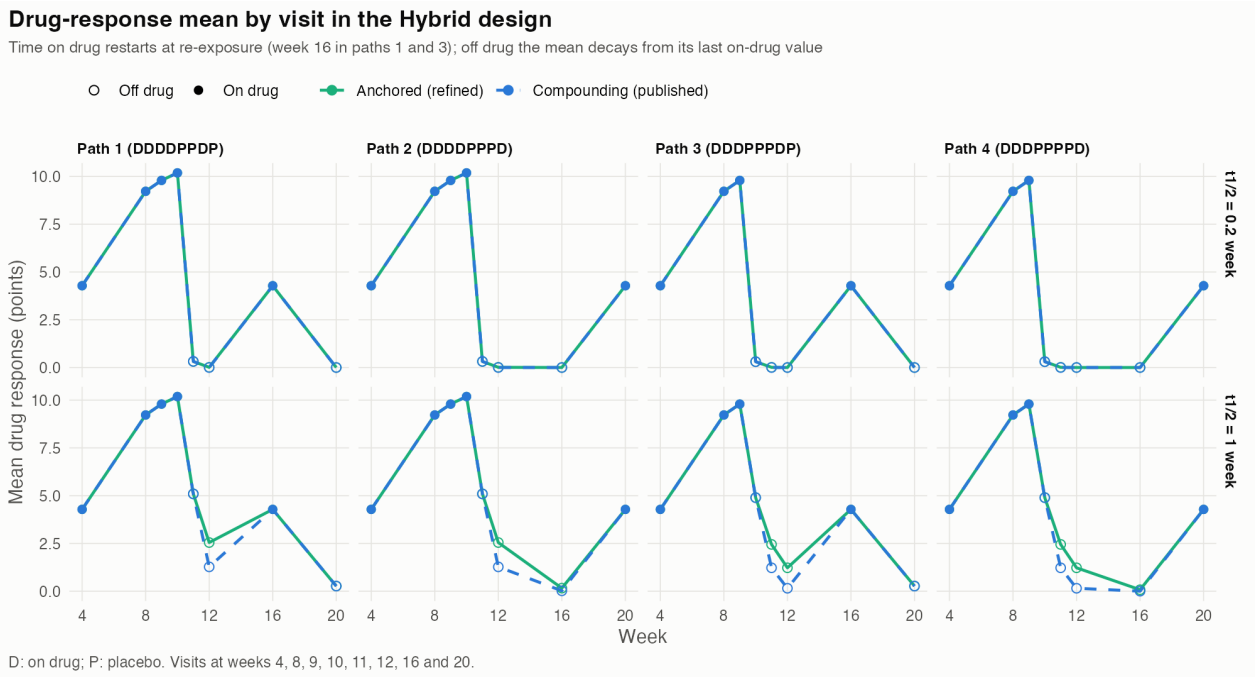



Figure 6: Drug-response mean by visit in the Hybrid design, by path, under the extended carryover rule (decay from the last on-drug mean, the extended DGP) and the original carryover rule (compounded decay), at carryover half-lives of 0.2 and 1 week. Filled points: on-drug visits; open points: off-drug visits. D: drug; P: placebo.

of time on drug at re-exposure, which for prazosin matches re-titration after an interruption. For drugs without titration the response should instead rebuild from its residual level. That is an option, not the default. The choice rests on fidelity to the pharmacology, as a judgment. The numerical consequence is at most 0.006 in power either way.

## Stage 4. Serial correlation

The original DGP uses compound symmetry. Visits one week apart and sixteen weeks apart have the same correlation of 0.8. We have no PTSD symptom series with which to decide whether that overstates long-lag correlation, and we do not claim that it does. Both compound symmetry and AR(1) are used in N-of-1 simulation (17). Serial correlation has been discussed as a design consideration for personalized studies (30). Two elements are open: the form of the within-component correlation, and its value.

The extended DGP uses AR(1) in calendar weeks, $\rho^{|w_t - w_s|}$, with one $\rho$ for all three components. The value of $\rho$ is a parameter. We take $\rho = 0.7$ as a convention, not an estimate. It is the original 0.8 lowered by 0.1, which makes the original effect size of 0.3 feasible under the original structure

Table 6: Serial correlation: the feasibility bound on $c_{bm}$ (Ceiling; decaying correlation, no carryover, minimum over the Hybrid, crossover and OL+BDC designs) and closed-form E9 power (Hybrid, $N = 70$, no carryover) by $\rho$, under compound symmetry and under the separable AR(1) form of Stage 5. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the separable form the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho \leq 0.7$, and the crossover design is never binding. At $t_{1/2} = 0.2$ week every power value is lower by 0.004 to 0.009.

| | Compound symmetry | | Separable AR(1) | |
|---|---|---|---|---|
| $\rho_{cs}$ or $\rho$ | Ceiling | Power at $c_{bm} = 0.25$ | Ceiling | Power at $c_{bm} = 0.25$ |
| 0.5 | 0.458 | 0.492 | 0.448 | 0.261 |
| 0.6 | 0.410 | 0.553 | 0.462 | 0.270 |
| 0.7 | 0.346 | 0.628 | 0.474 | 0.290 |
| 0.8 | 0.256 | 0.721 | 0.440 | 0.335 |

(below). It should be read with the sensitivity range in Table 6.

Table 6 shows what $\rho$ does under compound symmetry and under the structure adopted in Stage 5, for $\rho \in \{0.5, 0.6, 0.7, 0.8\}$. Under compound symmetry alone, lowering $\rho_{cs}$ from 0.8 to 0.7 raises the feasibility bound from 0.256 to 0.346. The original effect size of 0.3 then becomes feasible by the change of $\rho$ alone, with no change of structure. The same change lowers closed-form power at $c_{bm} = 0.25$ from 0.721 to 0.628. A lower serial correlation leaves more within-person noise in the on-minus-off contrast. The two criteria pull in opposite directions. The feasible range widens as $\rho$ falls, and the power at a matched $c_{bm}$ falls. The paper's own default should be read in that light.

The extended DGP uses AR(1) in weeks at $\rho = 0.7$. The choice rests on existence, traded against power. The calibration status is open. The AR(1) form is adopted for the reason given in Stage 5 (it is what the analysis model assumes), not because compound symmetry has been shown wrong for these data. Any study using the DGP should report $\rho$. Where the conclusion could depend on it, the analysis should be repeated at 0.6 and 0.8.

## Stage 5. Cross-component structure

The original DGP has a same-visit cross-component correlation $c_1 = 0.2$ and an other-visit correlation $c_\times = 0.1$ at every lag. With AR(1) within components, the direct translation keeps $c_1$ at lag zero and lets the other-visit correlation decay, $c_\times \rho^{|w_t - w_s|}$. This two-constant form is what the package's `buildSigma()` builds when its two constants differ (Appendix B.5). It is not the only choice, and it has a validity condition.

The extended DGP uses the separable form, $c_1\rho^{|w_t-w_s|}$ at every lag including zero. The response block is then a Kronecker product $M = K \otimes A$ of a $3 \times 3$ component correlation $K$ (1 on the diagonal, $c_1$ off it) and one AR(1) kernel $A_{ts} = \rho^{|w_t-w_s|}$. In words, every component has the same first-order autoregressive correlation, and the between-component correlation is constant at every lag. It is the two-constant form with $c_\times = c_1$. The original same-visit value $c_1 = 0.2$ is kept. Four properties follow (Appendix A.5).

1. **Validity for every schedule.** The eigenvalues of $M$ are products of those of $K$ and $A$, so $M$ is positive definite for any visit times. The two-constant form adds a same-visit cross-component term that carries correlation without variance. It is positive definite only when the smallest eigenvalue of $A$ exceeds $(c_1 - c_\times)/(1 - c_\times) = 0.111$. On the paper's schedules the condition holds (smallest eigenvalue 0.122 in the Hybrid and OL+BDC designs at $\rho = 0.8$, 0.193 at 0.7; 0.28 and 0.43 in the crossover design). On 20 weekly visits at $\rho = 0.8$ it holds by less than 0.001. On daily measurement over eight weeks it fails (0.016 at $\rho = 0.8$, 0.026 at 0.7). The two-constant form therefore cannot be used for the dense schedules that wearable or diary outcomes produce.
2. **The original structure is already of this kind in pieces.** It is the sum of a person-level and an occasion-level separable term, each valid. This is why the original matrices fail only through the biomarker row, never in the response block.
3. **An exact, interpretable feasibility bound.** The bound factors into a component term and a schedule term,

$$c^*_{bm} = \sqrt{1 - R^2}\,\left(g^\top A^{-1} g\right)^{-1/2}, \qquad R^2 = \frac{2c_1^2}{1 + c_1},$$

   where $R^2$ is the squared multiple correlation of the drug response on the other two components at one visit. The schedule term is a sum of per-step terms of the correlation profile. A profile that switches abruptly between closely spaced visits has a large per-step term. This is why designs with many on-off switches can carry only small effects.
4. **It matches the analysis model.** Under separability the summed response is an AR(1) process, up to the expectancy scaling of the placebo-response component. That is the residual structure `corCAR1` assumes. Under the two-constant form it is AR(1) plus a white-noise term the analysis model omits. This is derived. Its effect on test calibration has not been measured.

Table 7: Largest feasible biomarker-response correlation $c_{bm}$ under the decaying correlation, without carryover, minimum over the Hybrid, crossover and OL+BDC designs. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the two AR(1) forms the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho = 0.7$ (separable form: Hybrid 0.480), and the crossover design is never binding.

| $\rho_{cs}$ or $\rho$ | Compound symmetry | AR(1), two-constant cross-component | Separable AR(1) |
|---|---|---|---|
| 0.8 | 0.256 | 0.291 | 0.440 |
| 0.7 | 0.346 | 0.452 | 0.474 |

Table 7 gives the feasibility bounds. At the original $\rho = 0.8$ the two-constant form raises the bound only to 0.291. A single same-visit correlation repeated across visits that share little demands more of the biomarker row. In the open-label design, where every visit is on drug, it lowers the bound. The separable form raises the bound to 0.440. At $\rho = 0.7$ the two forms are close, 0.452 against 0.474. At the recommended $\rho$ the separable form's gain in feasible range is therefore small, and its reasons are properties 1 and 4. The bound of 0.474 is attained in the OL+BDC design (0.480 in the Hybrid design). A correlation of 0.45 is feasible in every design without carryover. Effects as large as the original 0.6 exceed every feasibility bound in the Hybrid design.

The wider feasible range has a consequence that does not depend on the form. At the same $c_{bm}$, AR(1) data carry less information about the interaction than compound-symmetric data. The closed-form power of the paired-difference statistic in the Hybrid design at $c_{bm} = 0.25$ falls from 0.721 under the original structure to 0.335 under the separable form at $\rho = 0.8$ and 0.290 at $\rho = 0.7$. A larger feasible correlation does not by itself mean more power. Effect sizes under different structures should therefore be compared on the scale of the implied slope, not of $c_{bm}$.

The extended DGP uses the separable form with $c_1 = 0.2$. The choice rests on analyzability (property 4) and existence (properties 1 and 3). The value $c_1 = 0.2$ is the original value and is uncalibrated.

## Stage 6. Features the extended DGP adds

The original DGP correlates the biomarker with the drug response only. A biomarker that is also prognostic, through an association with the natural history, the placebo response or baseline severity, cannot be represented. A simulation built on the original DGP therefore cannot show whether an analysis stays valid under such a biomarker.

The extended DGP adds optional correlations of the biomarker with the natural-history component or the placebo-response component (constant over visits, or growing along the natural-history curve) and with baseline severity. It adds phase-specific measurement error, a Weibull decay form and monotone dropout as further options.

**A prognostic biomarker.** We simulated a correlation of 0.3 between the biomarker and the natural-history component (constant, or growing along its curve), the placebo-response component, or baseline severity, with no interaction. Analyses that keep baseline as an outcome row use one biomarker main effect $\beta_B$ for baseline and for the off-drug visits. When the biomarker is prognostic through a pathway that is absent at baseline, those two associations differ. Part of the difference is attributed to the interaction (Appendix A.4). Every analysis that shares the biomarker effect with baseline exceeded the nominal level in at least one such cell (Table 8).

Table 8: Null rejection rates under a prognostic biomarker, 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell), first run of study 3 (original carryover rule for the mean, fixed allocation). The rerun with carryover decaying from the last on-drug mean changed no analysis's verdict. Rates above 0.075 in bold.

| Analysis | Natural history, constant | Natural history, growing | Placebo response | Baseline severity |
|---|---|---|---|---|
| Original analysis (random intercept, model-based) | **0.107** | 0.053 | **0.159** | 0.075 |
| Random intercept + CAR(1), CR2 | **0.099** | 0.049 | **0.100** | 0.050 |
| The same, half-life chosen by AIC | **0.115** | 0.058 | **0.112** | 0.055 |
| Unstructured covariance, Kenward-Roger | **0.416** | **0.195** | **0.222** | 0.051 |
| The same, half-life chosen by AIC | **0.509** | **0.245** | **0.256** | 0.064 |
| Random intercept + CAR(1) with expectancy terms, CR2 | 0.047 | 0.046 | 0.040 | 0.040 |
| The same, with phase-specific residual variances, CR2 | 0.055 | 0.044 | 0.042 | 0.042 |
| The same, with phase-specific residual variances, model-based | 0.062 | 0.043 | 0.045 | 0.046 |
| Expectancy terms, half-life chosen by AIC, CR2 | 0.059 | 0.052 | 0.047 | 0.045 |

| Analysis | Natural history, constant | Natural history, growing | Placebo response | Baseline severity |
|---|---|---|---|---|
| Random intercept + CAR(1) with free baseline slope, CR2 | 0.049 | 0.063 | 0.049 | 0.050 |

The analyses that free the biomarker's baseline association kept the nominal level in every prognostic cell. The free-baseline model was at the edge of the band (0.063) under the growing pathway, which Appendix A.4 explains. One family adds expectancy and a biomarker-by-expectancy term to the mean model, with or without residual variances that differ by trial phase. The other adds `bmc:bl0`, which gives baseline its own biomarker slope. A prognostic association with baseline severity alone did less harm, because baseline severity enters every visit equally. The original analysis still reached 0.075 there. The scenarios in which the pathway is absent at baseline are the ones that show the inflation. Whether such a pathway, at a magnitude of 0.3, is realistic for blood pressure in PTSD is not known. The point of the option is that a DGP which cannot generate it cannot show the inflation.

**Feasibility with a prognostic correlation.** A prognostic correlation occupies the same biomarker row of the correlation matrix as $c_{bm}$. The two therefore share one feasibility condition. Writing the row as a sum of fixed patterns scaled by the correlations, the existence condition of Stage 1 becomes a quadratic form. The largest value of one correlation given the others is the positive root of a quadratic (Appendix A.4). Table 9 gives the bounds for the Hybrid design under the extended DGP. The prognostic correlation of 0.3 used above lies inside the feasible region at every $c_{bm}$ up to 0.45. At the stress test's half-life of 0.5 week, a prognostic correlation of 0.3 with $c_{bm} = 0.45$ lies 0.006 inside the boundary. A prognostic correlation of 0.6 is infeasible with any $c_{bm}$. That is the combination the companion study's two repaired runs had simulated (Stage 1). The correlations with the placebo-response and the natural-history components have identical bounds, because their patterns are the same constant on every visit of one component. A correlation with baseline severity has a looser bound, since baseline severity is uncorrelated with the response components and enters only through its square. The OL+BDC design is within 0.03 of the Hybrid values. The crossover design has looser bounds throughout. We verified the bounds by checking that the smallest eigenvalue of the full matrix is positive at 0.999 times each bound

Table 9: Largest feasible prognostic correlation given the biomarker-response correlation, and the reverse, under the extended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$), Hybrid design, minimum over its four paths. Correlation: the quantity whose bound is given. $c_{PB}$, $c_{TV}$, $c_{BL}$: correlation of the biomarker with the placebo-response component, the natural-history component and baseline severity. None: no value is feasible.

| Correlation | Given | $t_{1/2} = 0.2$ week | $t_{1/2} = 0.5$ week |
|---|---|---|---|
| $c_{PB}$ or $c_{TV}$ | alone | 0.513 | 0.513 |
| | $c_{bm} = 0.25$ | 0.466 | 0.477 |
| | $c_{bm} = 0.30$ | 0.435 | 0.451 |
| | $c_{bm} = 0.45$ | 0.243 | 0.306 |
| $c_{BL}$ | alone | 1.000 | 1.000 |
| | $c_{bm} = 0.25$ | 0.857 | 0.876 |
| | $c_{bm} = 0.30$ | 0.786 | 0.815 |
| | $c_{bm} = 0.45$ | 0.374 | 0.496 |
| $c_{bm}$ | alone | 0.485 | 0.518 |
| | $c_{PB} = 0.3$ | 0.422 | 0.454 |
| | $c_{PB} = 0.6$ | none | none |

and negative at 1.001 times it.

**Size more generally.** Pooled over the 11 non-prognostic null cells (11,000 replicates; nominal band 0.046 to 0.054), the original analysis rejected 0.069, and 0.081 at $N = 35$. The unstructured model with Kenward-Roger inference (31) rejected 0.057, with 0.087 at $N = 35$. Random-intercept models with continuous-time AR(1) residuals and CR2 cluster-robust standard errors (32,33) rejected 0.048 to 0.053, with no single cell above 0.069. The expectancy-term model with its half-life chosen by AIC rejected 0.057, with 0.073 when the true half-life was one week. Both values of 0.057 lie above the pooled band.

**The options.** Table 10 shows what each optional feature did to size and power for three analyses: the original analysis, the random-intercept CAR(1) model with CR2, and the expectancy-term model that the prognostic cells favor. Measurement error (SD 3 open label, 6 blinded) lowered power by 0.05 to 0.11 and left size within its band. Dropout of 20%, whether completely at random or dependent on the last score, lowered power by 0.02 to 0.07. The original analysis reached 0.079 under random dropout. A Weibull decay with the same half-life lowered power by 0.09 to 0.12 at shape 0.5 (a steep early decay) and raised it by 0.03 to 0.07 at shape 2. The true half-life itself moved power from 0.48 to 0.54 at zero to 0.31 to 0.39 at one week. Halving $N$ roughly halved power. Doubling it brought the CAR(1) model to 0.82. None of these features changed which analyses are valid, which is the analyzability criterion. They change power, which the planning section and the

Table 10: The optional features of the extended DGP: null rejection rate ($c_{bm} = 0$; 1,000 replicates, MCSE 0.007) and power ($c_{bm} = 0.30$; 500 replicates, MCSE at most 0.022) for the original analysis (Original), the random-intercept CAR(1) model with CR2 and the expectancy-term model with CR2 (Expect.). Hybrid design, first run of study 3. Each row changes one feature of the base cell.

| | Null rejection | | | Power | | |
|---|---|---|---|---|---|---|
| Cell | Original | CAR(1) | Expect. | Original | CAR(1) | Expect. |
| Base cell ($N = 70$, $t_{1/2} = 0.5$, exponential, no dropout) | 0.076 | 0.049 | 0.062 | 0.476 | 0.508 | 0.410 |
| Measurement error (SD 3 and 6) | 0.071 | 0.051 | 0.047 | 0.424 | 0.400 | 0.310 |
| Dropout 20 percent, completely at random | 0.079 | 0.059 | 0.051 | 0.460 | 0.466 | 0.382 |
| Dropout 20 percent, dependent on last score | 0.064 | 0.056 | 0.044 | 0.426 | 0.436 | 0.366 |
| Weibull decay, shape 0.5 | 0.073 | 0.047 | 0.046 | 0.384 | 0.384 | 0.298 |
| Weibull decay, shape 2 | 0.066 | 0.037 | 0.045 | 0.538 | 0.536 | 0.476 |
| $t_{1/2} = 0$ | 0.072 | 0.036 | 0.049 | 0.540 | 0.538 | 0.484 |
| $t_{1/2} = 1$ week | 0.061 | 0.043 | 0.061 | 0.388 | 0.384 | 0.310 |
| $N = 35$ | 0.081 | 0.058 | 0.053 | 0.270 | 0.238 | 0.206 |
| $N = 150$ | 0.063 | 0.047 | 0.047 | 0.770 | 0.820 | 0.726 |

companion paper address.

**Power and robustness.** The expectancy-term model keeps its size under every prognostic pathway and has lower power: 0.410 against 0.508 for the CAR(1) model in the base cell (first run), and 0.431 against 0.496 in size-adjusted power pooled over the reruns, about 13% less. Under compound symmetry at $c_{bm} = 0.25$ the order reverses (0.510 against 0.490). A DGP paper can say which analyses are valid under which features. The choice between power and robustness is the trialist's. It depends on whether a prognostic pathway absent at baseline is plausible for the biomarker in hand.

The extended DGP includes the prognostic correlations as an option. A simulation that omits them cannot show the size inflation above. The other three features are options whose effect on power is tabulated. The choice rests on analyzability. Measurement error is implemented in the stress-test driver and not yet in the package (Appendix B.5).

## Stage 7. The extended DGP

Table 11 collects the choices. Each row is marked as an extension of the original DGP, a choice among open alternatives, an option, or a retained element, and names the stage that justifies it. Values not listed are the original values.

Table 11: The extended data-generating process. Defaults are those used in the stress test unless stated.

| Element | Specification | Default | Kind | Justified in |
|---|---|---|---|---|
| Components | $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, Gompertz means as in the original DGP | original parameters | retained | Methods |
| Validity | stop if the correlation matrix is not positive definite | always | extension | Stage 1 |
| Biomarker-response correlation | $c_{bm}$ on drug; $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure | $c_{bm}$ below the feasibility bound of every design in the study | extension | Stage 2 |
| Carryover on the mean | decay from the last on-drug value, $\mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$ | $t_{1/2} = 0.1$ to 0.2 week (pharmacological, assumed); 0.5 in the stress test | choice | Stage 3 |
| Re-exposure | as in the original DGP: time on drug restarts at zero | on, for titrated drugs | retained | Stage 3 |
| Within-component correlation | AR(1) in weeks, $\rho^{\lvert w_t - w_s \rvert}$, one $\rho$ for all three components | $\rho = 0.7$ (convention; report and vary) | choice | Stage 4 |
| Cross-component correlation | separable, $c_1\rho^{\lvert w_t - w_s \rvert}$ at every lag; the Kronecker form applies to the correlation, since the PB SD scales with the expectancy | $c_1 = 0.2$ (original value) | choice | Stage 5 |
| Path allocation | fixed numbers, as in the original DGP | original values | retained | Methods |

| Element | Specification | Default | Kind | Justified in |
|---|---|---|---|---|
| Prognostic correlations | correlation of the biomarker with TV or PB, or with baseline severity; or a mean shift of TV growing along its curve | off; 0.3 in the stress test; correlations must lie inside the joint bound with $c_{bm}$ (Table 9) | option | Stage 6 |
| Measurement error | independent, SD by phase | off; 3 open label, 6 blinded in the stress test | option (driver only) | Stage 6 |
| Decay form | exponential or Weibull with the same half-life | exponential | option | Stage 6 |
| Dropout | monotone, completely at random or at random given the last score | off; 20% in the stress test | option | Stage 6 |

## Using the extended DGP to plan a trial

**What $c_{bm}$ means.** Conditional on the biomarker, the correlation encoding shifts the on-drug drug-response mean by $c_{bm}\sigma_{BR}$ per standard deviation of biomarker (Discussion). With the original $\sigma_{BR} = 8$, $c_{bm} = 0.25$ is a slope of 2.0 symptom points per SD of biomarker, 0.35 is 2.8 points, 0.45 is 3.6 points, and the feasibility bound of 0.474 is 3.8 points. The biomarker's SD is 15.4 mm Hg in the original parameters, so 0.25 is 0.13 points per mm Hg. The observed association for prazosin is 14 points on the Clinician-Administered PTSD Scale per 10 mm Hg (9), or 21.5 points per SD of biomarker. Representing that slope under the correlation encoding would need $\sigma_{BR} \geq 45$ at the bound. That is larger than the SD of the symptom score itself (18.5 at baseline in the original parameters). Under the original parameters the bound is therefore below the prazosin association. The comparison is approximate for two reasons. The observed slope is on the total symptom change, not the drug-response component. And $\sigma_{BR} = 8$ is the original choice, not an estimate. A study that needs a slope above the bound should use the sensitivity model of the Discussion once it is

implemented, or raise $\sigma_{BR}$ with a stated reason.

**Sample size.** Table 12 gives the total $N$ for 80% power of the paired-difference statistic E9 under the extended DGP, by design, $c_{bm}$ and half-life, and the power at the original $N = 70$. The sample sizes are closed-form values for E9. In the Hybrid design E9 tracks the original mixed model closely (Methods). In the crossover design it understates a time-adjusted mixed model. At the pharmacological half-life the Hybrid design needs about 280 participants at $c_{bm} = 0.25$ and 84 at 0.45. At a 1-week half-life those rise to 480 and 146. For comparison, a main-effect test of the same standardized size would need about a quarter of the sample (1).

**The recommended analysis at the planned sample sizes.** The companion paper (29) recommends the analysis G9 for the interaction. G9 is a linear mixed model with a random intercept and continuous-time AR(1) residual correlation. Its drug indicator is the exposure-decayed $D_{bc}$ at an analysis half-life chosen by AIC over 0.25, 0.5, 1 and 2 weeks, with the interaction in the selection model. The chosen model is refitted by REML and the interaction is tested with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom. We simulated G9 under the extended DGP in every cell of Table 12, at $N = 70$ and at the E9 sample size for 80 percent power, with 1,000 replicates per cell. We also simulated the null ($c_{bm} = 0$) at $N = 70$ for each design and half-life with 2,000 replicates. The last column of Table 12 gives G9 power at $N = 70$ at the 0.2-week half-life, and Table 13 gives every cell. Where the E9 sample size exceeded 400, or was above 800 and not computed, the cell ran at $N = 400$ and is marked.

Three results follow. First, G9 rejected the null at 0.045 to 0.059 across the nine design-by-half-life cells (Monte Carlo SE at most 0.005). Second, at $N = 70$ G9 has higher power than E9 in every cell. In the Hybrid design at $c_{bm} = 0.25$ and the 0.2-week half-life, G9 power is 0.650 (MCSE 0.015) against 0.285 for E9. In the crossover design at the same cell it is 0.413 (MCSE 0.016) against 0.202. Third, at the E9 sample size for 80 percent power, G9 power is 0.984 to 1.000 in the 20 uncapped cells, and 0.899 to 0.999 in the 7 cells that ran at $N = 400$. The closed-form E9 rule is therefore conservative for G9. A trial sized by Table 12 and analyzed with G9 had power of at least 0.899 in every cell simulated. The sample size at which G9 reaches 80 percent power is smaller than the E9 value, and we did not search for it. Fits failed to converge in at most 0.013 of replicates in any cell, and the rejection rates are computed over converged fits.

**Parameters by status.**

Table 12: Planning under the extended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean): total $N$ for 80 percent power of the paired-difference statistic E9 at $\alpha = 0.05$, by design, interaction strength and carryover half-life, with the implied slope in symptom points per SD of biomarker ($\sigma_{BR} = 8$) and the power at $N = 70$ and the 0.2-week half-life for E9 (closed form) and for the analysis G9 (simulated, 1,000 replicates per cell, Monte Carlo SE at most 0.016). Participants are allocated to paths in the original fixed proportions.

| | | | N for 80 percent power | | | Power at $N = 70$, 0.2 week | |
|---|---|---|---|---|---|---|---|
| Design | $c_{bm}$ | Slope (points per SD) | $t_{1/2} = 0$ | 0.2 week | 1 week | E9 | G9 |
| Hybrid | 0.25 | 2.0 | 272 | 278 | 480 | 0.285 | 0.650 |
| | 0.35 | 2.8 | 138 | 140 | 244 | 0.503 | 0.919 |
| | 0.45 | 3.6 | 82 | 84 | 146 | 0.728 | 0.994 |
| OL+BDC | 0.25 | 2.0 | 394 | 404 | $> 800$ | 0.210 | 0.531 |
| | 0.35 | 2.8 | 200 | 206 | 452 | 0.368 | 0.823 |
| | 0.45 | 3.6 | 120 | 124 | 272 | 0.558 | 0.957 |
| Crossover | 0.25 | 2.0 | 424 | 424 | 454 | 0.202 | 0.413 |
| | 0.35 | 2.8 | 216 | 216 | 230 | 0.353 | 0.626 |
| | 0.45 | 3.6 | 130 | 130 | 138 | 0.537 | 0.867 |

Table 13: Rejection rate of the analysis G9 under the extended DGP of the preceding planning table, by design, interaction strength and carryover half-life, at $N = 70$ and at the total $N$ the closed-form E9 rule gives for 80 percent power (the preceding table). Rows with $c_{bm} = 0$ give the size at $N = 70$ (2,000 replicates per cell, Monte Carlo SE at most 0.005); the other rows give power (1,000 replicates per cell, Monte Carlo SE at most 0.016). †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Rejection rates are over converged fits (at most 1.3 percent of replicates failed in any cell).

| | | At $N = 70$ | | | At the E9 $N$ for 80 percent power | | |
|---|---|---|---|---|---|---|---|
| Design | $c_{bm}$ | $t_{1/2} = 0$ | 0.2 week | 1 week | $t_{1/2} = 0$ | 0.2 week | 1 week |
| Hybrid | 0.00 | 0.045 | 0.051 | 0.051 | | | |
| | 0.25 | 0.663 | 0.650 | 0.418 | 0.995 | 0.999 | 0.992† |
| | 0.35 | 0.919 | 0.919 | 0.669 | 0.996 | 1.000 | 0.997 |
| | 0.45 | 0.997 | 0.994 | 0.854 | 0.998 | 0.993 | 0.996 |
| OL+BDC | 0.00 | 0.053 | 0.051 | 0.055 | | | |
| | 0.25 | 0.559 | 0.531 | 0.251 | 1.000 | 0.999† | 0.899† |
| | 0.35 | 0.828 | 0.823 | 0.457 | 1.000 | 0.999 | 0.995† |
| | 0.45 | 0.978 | 0.957 | 0.648 | 1.000 | 0.999 | 0.996 |
| Crossover | 0.00 | 0.059 | 0.048 | 0.049 | | | |
| | 0.25 | 0.351 | 0.413 | 0.387 | 0.987† | 0.983† | 0.971† |
| | 0.35 | 0.651 | 0.626 | 0.647 | 0.984 | 0.989 | 0.985 |
| | 0.45 | 0.874 | 0.867 | 0.828 | 0.993 | 0.988 | 0.984 |

- *Original values:* the Gompertz parameters of the three components and their standard deviations; the biomarker and baseline means and standard deviations; $c_1 = 0.2$; the path allocation; the expectancy values; the restart of time on drug at re-exposure.
- *Chosen here:* the decaying correlation and the stop on infeasible matrices (the extensions); carryover decaying from the last on-drug mean; AR(1) in weeks with $\rho = 0.7$; the separable cross-component form; the symptom half-life range of 0.1 to 0.2 week; the option defaults used in the stress test.
- *Uncalibrated:* every value above. No parameter has been estimated from prazosin trial data. The ones to which the results are most sensitive are $\rho$ (Stage 4), $\sigma_{BR}$ (which sets the slope a given $c_{bm}$ implies) and the half-life.

**Choosing $c_{bm}$.** Take the smallest feasibility bound over the designs and half-lives in the study (Table 7 and Appendix A.5; 0.474 for the three designs at $\rho = 0.7$ without carryover, higher with carryover). State $c_{bm}$ as a slope in outcome points per SD of biomarker. Keep it below that bound. The package reports an infeasible value by stopping.

# Discussion

## The correlation encoding of the interaction

**Conditional structure.** Under joint normality, conditioning on the biomarker shows what the correlation encoding does to the drug response. With $b_i$ the standardized biomarker,

$$E(BR_{it} \mid B_i) = \mu_{BR,t} + c_{bm}\, g_t\, \sigma_{BR}\, b_i, \qquad \mathrm{Cov}(BR_{i\cdot} \mid B_i) = \Sigma_{BR} - c_{bm}^2 \sigma_{BR}^2\, g g^\top.$$

The encoding therefore shifts the conditional drug-response mean in proportion to the correlation profile $g_t$. It reduces the residual drug-response covariance by a rank-one term. Two consequences follow.

- **The power loss under carryover is set by the profile.** The size of the loss depends on whether the interaction fades with the drug effect, which is the correlation rule.
- **The feasibility bound is the condition** that the reduced residual covariance remains positive definite.

The contribution of the original DGP is therefore a parameterization of the interaction as a correlation that keeps the marginal variance of the drug response fixed. That parameterization was new in the N-of-1 literature, to our knowledge. Both its strengths and its limits follow from it.

The correlation encoding has four strengths.

- **One interpretable parameter.** The interaction strength is a single correlation. Its square, $c_{bm}^2$, is the share of on-drug drug-response variance the biomarker explains.
- **A reduced form of responder heterogeneity.** Suppose participants fall into two latent responder classes, and both the biomarker and the drug response differ between classes. Then class membership carries the covariance between them. Matching second moments gives $c_{bm}^2 = f_B f_{BR}$, where $f_B$ and $f_{BR}$ are the fractions of the biomarker's and the drug response's variance that lie between classes (Appendix A.6). The encoding therefore matches the second moments of a responder-class mechanism without requiring the mixture to be identified. Identifying the mixture typically needs larger samples than N-of-1 programs enroll (34). The match is limited to second moments. The mixture's conditional mean is nonlinear in the biomarker, and its higher moments differ.
- **The marginal variance of the drug response is fixed.** The biomarker explains part of the existing variation rather than adding to it. The effect size can therefore be stated as a share of response variance without changing the response distribution.
- **It fits a biomarker that marks individual sensitivity to the drug** better than one that sets a fixed dose-response shift.

Its limits, all made explicit by this analysis, are structural.

- **The joint distribution must exist,** which bounds the effect size. In outcome units the bound is $c_{bm}^* \sigma_{BR}$ per SD of biomarker, 3.8 points under the extended DGP at the original $\sigma_{BR}$. The planning section shows this is below the observed prazosin association. The bound depends on the correlation structure and the design.
- **The correlation must be specified at off-drug visits.** A constant correlation after first exposure removes the interaction under carryover. The specification adopted here is itself asymmetric between titration and washout (Stage 2).
- **The drug-response component keeps its full variance at off-drug visits,** with zero mean. This adds noise to every on-off contrast. It belongs to the component decomposition,

not to the correlation encoding.

- **A correlation is less intuitive than a slope for clinical readers.** The slope it implies, $-c_{bm}\sigma_{BR}/\sigma_{bm}$, must be derived before it can be compared with an observed effect. The comparison depends on $\sigma_{BR}$, which is not estimated.

A natural next step addresses these limits. Generate the drug response from a participant-level sensitivity to exposure,

$$BR_{it} = \omega_i f(t) + \varepsilon_{it}, \qquad \omega_i = \bar{\omega}\,(1 + \zeta b_i) + u_i,$$

where $f(t)$ is the exposure-response time course, $\zeta$ the biomarker's effect on sensitivity and $u_i$ the sensitivity it does not explain. This is a random-coefficient model with a covariate-by-exposure interaction, the form used to describe individual response in replicated crossover and N-of-1 trials (7,20).

- It encodes the biomarker through individual sensitivity, so it has no feasibility bound.
- Its moderation is proportional to the response at every visit, so it fades during titration as well as after discontinuation.
- Its off-drug variance shrinks with the response.

It is not yet implemented in the software. Its response half-life for prazosin in PTSD is not known. Until then, the correlation encoding's range of use is effect sizes below the feasibility bound, which the planning section translates into outcome units.

## What remains uncalibrated

The extended DGP meets the existence, identifiability and analyzability criteria by construction and derivation. It meets the fidelity criterion by judgment. It does not meet the calibration criterion: every parameter is the original value or a choice made here. The values that matter most are the serial correlation, the drug-response SD and the symptom half-life. All three could be estimated from an aggregated N-of-1 trial of prazosin with weekly symptom measurement. Such a trial would also settle whether the compound-symmetry or the AR(1) form fits such series.

## Recommendations

For simulation studies that build on the original DGP, we recommend the following.

1. **Apply both extensions:** stop on non-positive-definite matrices, and let the biomarker-response correlation decay after discontinuation.
2. **State the choices:** carryover decaying from the last on-drug mean, AR(1) in weeks at a reported $\rho$, and the separable cross-component form. Repeat the analysis at neighboring $\rho$ where the conclusion could depend on it.
3. **Choose $c_{bm}$ below the feasibility bound** for every design in the study. Report it on two scales: as $c_{bm}$, and as the implied slope in outcome points per SD of biomarker.
4. **Keep the original DGP as a reference arm** whenever results are compared with the original ones. Attribute each difference to a named stage.
5. **Include a prognostic scenario** whenever the analysis shares the biomarker effect with the baseline visit. State the trade-off between the power of the plain model and the robustness of the expectancy-term model rather than resolving it for the trialist.
6. **Use the sensitivity model, once available, for effects above the feasibility bound.** Until then, state the bound in outcome units as a limitation.

## Limitations

- **The closed form covers one statistic.** It is exact for the paired-difference statistic, not for the original mixed model. The two agree closely in the Hybrid design. In the crossover design the closed form understates what a time-adjusted analysis achieves. The sample sizes of the planning table inherit this. The simulated G9 column shows the direction and size of the difference: at the E9 sample size G9 power was 0.98 to 1.00 in the uncapped cells. The G9 sample size for 80 percent power was not searched for.
- **At pharmacological half-lives the extended DGP shows no carryover effect.** The on/off correlation difference is within 0.01 of 1 at 0.2 week. The DGP therefore tests carryover only at half-lives of 0.5 week and above, which we cannot ground pharmacologically.
- **The correlation is asymmetric between titration and washout.** Full moderation during the first weeks on drug is a modeling choice inherited from the original DGP, not a consequence of the pharmacology.

- **Parameter values are not estimated.** The Gompertz parameters, correlations, $\sigma_{BR}$ and prognostic correlations are original values or chosen values, not estimates from prazosin trial data.
- **The DGP is Gaussian with a single biomarker.**
- **The pharmacological carryover range** for prazosin at weekly visits is inferred from its plasma half-life, not measured as a symptom-response half-life.
- **The stress test is exploratory.** Cells and analyses were added after first results. Its prognostic scenarios are the ones that show the size inflation by construction.

# Conclusions

The original DGP encodes the predictive-biomarker interaction as a correlation between the biomarker and the drug response. That encoding remains a coherent choice for simulating aggregated N-of-1 trials. The original DGP holds the correlation constant after first exposure, so the interaction is absent whenever there is carryover. It repairs correlation matrices that are not positive definite. Both features have closed-form explanations, and both are changed by small extensions. With a correlation that decays after discontinuation and a feasibility check that stops instead of repairing, the extended DGP returns the no-carryover power at pharmacological half-lives (the on/off correlation difference is within 0.01 of 1 at 0.2 week). Power declines smoothly at half-lives of 0.5 week and above, and monotonically for the paired-difference statistic. A serial correlation of 0.7 and a separable AR(1) structure allow a biomarker-response correlation of 0.47, or 3.8 symptom points per SD of biomarker at the original drug-response SD. The planning table quantifies the power at matched strength, for the paired-difference statistic in closed form and for the recommended mixed-model analysis G9 by simulation. At the closed-form sample size, G9 power was 0.98 to 1.00 in every uncapped cell. We propose the extended DGP, with options for prognostic pathways and other features that change which analysis is valid, as the reference for simulation studies of these designs. Every parameter is labeled as original, chosen or uncalibrated.

# Glossary

Terms are listed in the order in which the paper introduces them. Labels in backticks are the identifiers used in the repository's code, data files and table headings.

- **Original DGP** (`published`, "Published" and "Step" in tables): the data-generating process of Hendrickson et al. (2020), as implemented in their code at commit 58b32a9.
- **Extended DGP** (`refined`, "Refined" in tables): the data-generating process of Table 11, which adds the decaying correlation, the feasibility check, carryover decaying from the last on-drug mean, the separable AR(1) structure and the optional features to the original DGP.
- **Correlation encoding of the interaction** (`mvn`): the predictive biomarker is a correlation between the biomarker and the drug response that is present on drug and absent off drug.
- **Biomarker-response correlation** ($c_{bm}$, `c.bm`): the correlation between the baseline biomarker and the drug-response component at an on-drug visit, which sets the strength of the interaction.
- **Correlation profile** ($g_t$): the fraction of $c_{bm}$ present at visit $t$, equal to 1 on drug.
- **Original rule** (`step`, "Step" in tables): the biomarker-response correlation is $c_{bm}$ at every visit where the drug-response mean is nonzero, and so is constant after first exposure.
- **Decaying correlation** (`graded`, "Graded" in tables): the extended rule, under which the correlation is $c_{bm}$ on drug, $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before exposure.
- **On/off correlation difference** ($\mathcal{G}$): the on-drug value of $g_t$ minus the mean of $g_t$ over the post-baseline off-drug visits, $1 - \bar{g}_{\text{off}}$, through which alone the correlation rule affects closed-form power.
- **Feasibility bound** ($c^*_{bm}$, "Ceiling" in tables): the largest $c_{bm}$ for which the correlation matrix is positive definite under a given correlation structure, design and correlation profile.
- **Carryover half-life** ($t_{1/2}$, `carryover_t1half`): the time after discontinuation for the drug-response mean, and under the extended rule the biomarker-response correlation, to fall by half.
- **Pharmacological and behavioral half-lives**: half-lives of 0.1 to 0.2 week are the assumed pharmacological range for prazosin, and half-lives of 0.5 week and above are labeled behavioral scenarios.
- **Original carryover rule** (`recursive`): the drug-response mean at an off-drug visit is the previous visit's mean decayed by the time since discontinuation, so the decay compounds across visits.
- **Extended carryover rule** (`anchored`): the drug-response mean at an off-drug visit is the last on-drug mean decayed by the elapsed time since discontinuation.
- **Drug-response component** (BR): the latent response driven by time on drug, with mean

on a modified Gompertz curve.

- **Placebo-response component** (PB): the latent response driven by time in the trial and scaled by the expectancy.
- **Natural-history component** (TV): the latent response driven by time in the trial that does not depend on treatment.
- **Baseline severity** (BL): the participant's symptom level before any response, which enters every visit equally.
- **Expectancy** ($\eta_t$): the participant's probability of being on drug at visit $t$ given the design, 1 in the open-label phase and 0.5 when blinded.
- **Compound symmetry** ($\rho_{cs}$): the same correlation between any two visits of a component, whatever their separation.
- **Separable AR(1) structure**: the same first-order autoregressive correlation $\rho^{|w_t - w_s|}$ for every component, with a constant between-component correlation $c_1$ at every lag, so that the response block is a Kronecker product.
- **Two-constant form**: an AR(1) structure in which the same-visit cross-component correlation $c_1$ and the other-visit correlation $c_\times \rho^{|w_t - w_s|}$ use different constants.
- **Hybrid design** (`Hybrid`): open-label titration, blinded discontinuation, then a brief crossover, with visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20.
- **OL+BDC** (`OLBDC`): open label followed by blinded discontinuation.
- **CO**: the crossover design, in which participants receive drug and placebo in one of two orders.
- **OL**: the open-label design, in which every visit is on drug.
- **Path**: one sequence of drug and placebo phases within a design, to which participants are allocated in fixed numbers.
- **Paired-difference statistic** (E9, E9b): the per-participant difference between the mean on-drug and mean off-drug scores, regressed on the biomarker, using post-baseline visits only (E9) or counting baseline as one more off-drug visit (E9b).
- **Original analysis** ("Published" in tables): the linear mixed model of the original study, with a random intercept, a binary drug indicator and baseline included as an off-drug row.
- **G9** (`E7cr2` in data labels): the analysis the companion paper recommends; a random-intercept CAR(1) mixed model with the exposure-decayed drug indicator at a half-life chosen by AIC over 0.25, 0.5, 1 and 2 weeks, refitted by REML, with CR2 cluster-robust standard errors and Satterthwaite degrees of freedom.

- **Capped cell** († in Table 13): a cell whose E9 sample size for 80 percent power exceeds 400, simulated at $N = 400$.
- **Exposure-decayed drug indicator** ($D_{bc}$): a drug indicator equal to 1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after exposure, and 0 before exposure.
- **Prognostic correlation** ($c_{PB}$, $c_{TV}$, $c_{BL}$; `prog-PB`, `prog-TV-const`, `prog-TV-grow`, `prog-BL` in data labels): a correlation of the biomarker with the placebo-response component, with the natural-history component (constant over visits, or growing along its curve), or with baseline severity.
- **Expectancy-term model**: the random-intercept CAR(1) analysis that adds the expectancy and a biomarker-by-expectancy term to the mean model.
- **Free-baseline model**: the random-intercept CAR(1) analysis that adds `bmc:bl0`, a separate biomarker slope for the baseline row.
- **CAR(1)**: the continuous-time first-order autoregressive residual correlation `corCAR1` of the analysis model.
- **Calibration ratio** ($\kappa$): the mean model standard error divided by the empirical standard deviation of the estimates.
- **Stress test** (`11-strawman-stress-test.R`): study 3, the simulation that fits fourteen analyses to trials from the extended DGP with and without a prognostic biomarker.
- **The five criteria**: existence (the joint distribution exists), identifiability under carryover (the interaction remains an on-minus-off contrast), fidelity to the pharmacology, analyzability (the analysis model can represent the generated structure and the DGP can generate the features that decide validity), and calibration status (each parameter is labeled original, chosen or uncalibrated).

# Abbreviations

ADEMP: aims, data-generating mechanisms, estimands, methods, performance measures; AIC: Akaike information criterion; BDC: blinded discontinuation; BL: baseline severity; BR: drug-response component; CAPS: Clinician-Administered PTSD Scale; CAR(1): continuous-time first-order autoregressive correlation; CO: crossover; CR2: bias-reduced cluster-robust variance estimator; CS: compound symmetry; DGP: data-generating process; E9, E9b: the paired-difference interaction statistic on post-baseline visits, and with baseline counted as an off-drug visit; MCSE:

Monte Carlo standard error; OL: open label; OL+BDC: open label followed by blinded discontinuation; PB: placebo-response component; PD: positive definite; PTSD: post-traumatic stress disorder; TV: natural-history component.

# Declarations

**Ethics approval and consent to participate.** Not applicable: the study uses simulated data only.

**Consent for publication.** Not applicable.

**Availability of data and materials.** All code and simulated data are in the pmsimstats-ng repository (`https://github.com/rgt47/pmsimstats-ng`); a tagged release with a persistent identifier will be deposited at submission, and the commit reported here should be cited until then.

- the closed-form calculations, in `analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-fo` and the serial-correlation stage, planning table and trajectory figure, in `17-rho-stage.R` in the same directory, and the joint feasibility bounds with a prognostic correlation in `18-prognostic-joint-ceiling.R`;
- the G9 simulation at the planned sample sizes and on the Figure 4B grid, in `19-g9-planning-and-fig4b.R` in the same directory, with its outputs under `analysis/data/quick-sim/carryover-closed-form/g9-plan` and the combined three-version heatmap drawn by `21-fig4b-g9-heatmap.R`;
- the check of the original code, in `analysis/scripts/quick-sim/hendrickson-problems/14-graded-patch` and the repair counts in `11-pd-repair-count.R`;
- the stress test, in `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R`;
- the original code, at `github.com/rchendrickson/pmsimstats`, commit 58b32a9.

**Competing interests.** To be completed by the authors. Any author of this paper who is also an author of the original study must be disclosed here. The manuscript should state whether the original authors were consulted on the two extensions that change the original results.

**Funding.** To be completed by the authors.

**Authors' contributions.** To be completed by the authors.

**Acknowledgements.** To be completed by the authors.

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# Appendix A. Closed-form derivations

## A.1 Moments of the paired-difference statistic

Let $a_t = 1/n_{\text{on}}$ at on-drug visits and $-1/n_{\text{off}}$ at off-drug visits, so that $\Delta_i = \sum_t a_t Y_{it}$. Because the weights sum to zero, the baseline level cancels and $\Delta = -\sum_t a_t S_t$, with $S_t = TV_t + PB_t + BR_t$. In general $\text{Var}(\Delta) = a^\top \Sigma_S a$, where $\Sigma_S$ is the covariance of the summed response, whatever the structure.

With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_\eta = \sum a_t \eta_t$, $m_{\eta\eta} = \sum a_t^2 \eta_t^2$ and $m_{2\eta} = \sum a_t^2 \eta_t$, with $\eta_t$ the design expectancy, the variance under compound symmetry with common correlation $\rho_{cs}$ is

$$\begin{aligned}\text{Var}(\Delta) = {} & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho_{cs} m_1^2 + (1-\rho_{cs}) m_2] + \sigma_{PB}^2 [\rho_{cs} m_\eta^2 + (1-\rho_{cs}) m_{\eta\eta}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_\times m_1^2 + (c_1 - c_\times) m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_\times m_1 m_\eta + (c_1 - c_\times) m_{2\eta}],\end{aligned}$$

where $c_1$ and $c_\times$ are the same-visit and other-visit cross-component correlations. Under the separable form of Stage 5, $\text{Cov}(S_t, S_s) = (s_t^\top K s_s) A_{ts}$ (A.5), and $\text{Var}(\Delta) = \sum_{t,s} a_t a_s (s_t^\top K s_s) A_{ts}$, with $s_t$ the vector of component standard deviations at visit $t$.

- **Proposition 1.** No term depends on $t_{1/2}$ under either structure, so $\text{Var}(\Delta)$ is free of the half-life. The argument applies to any correlation matrix in which the half-life does not enter the response block. It therefore holds for unrepaired matrices only. The repair changes the response block, and its changes depend on the half-life through the correlation profile. For E9 the weights sum to zero, so $m_1 = 0$. For E9b the post-baseline weights sum to $1/(n_{\text{off}} + 1)$, and the baseline row, with $S_0 = 0$, carries the remaining weight.
- **Proposition 2.** Only $BR_t$ is correlated with $B$, with covariance $c_{bm} g_t \sigma_{bm} \sigma_{BR}$, so $\beta_{bm:D} = -c_{bm}(\sigma_{BR}/\sigma_{bm}) \sum_t a_t g_t$. With $g_t = 1$ on drug, this gives the expressions in the Methods.
- **Residual variance.** Under the correlation encoding, $\sigma_\Delta^2 = \text{Var}(\Delta) - \beta_{bm:D}^2 \sigma_{bm}^2$.

## A.2 Power

Within a path of $n$ participants, the least-squares slope is exactly normal given the biomarker values. Its $t$-statistic is exactly noncentral $t$ with $n-2$ degrees of freedom and noncentrality $\beta_{bm:D}\sqrt{\text{SS}_B}/\sigma_\Delta$,

where $\mathrm{SS}_B \sim \sigma^2_{bm}\chi^2_{n-1}$.

At $\mathrm{SS}_B = \sigma^2_{bm}(n-1)$ the noncentrality is $\Lambda = \sqrt{n-1}\, r_{\Delta B}/\sqrt{1-r^2_{\Delta B}}$, where $r_{\Delta B} = \beta_{bm:D}\sigma_{bm}/\mathrm{SD}(\Delta)$ is proportional to the on/off correlation difference. For small $r_{\Delta B}$ the noncentrality is approximately $\sqrt{n-1}\, r_{\Delta B}$. The sample size for a fixed noncentrality therefore scales as $1/r^2_{\Delta B}$ and hence approximately as $1/\mathcal{G}^2$. The exact scaling uses $r/\sqrt{1-r^2}$.

Pooling paths with a common intercept adds the between-path variation in mean contrast to the residual variance. In the Hybrid design it is under 0.7% of the total. Power is the noncentral-$t$ tail probability averaged over the $\chi^2$ law of $\mathrm{SS}_B$.

## A.3 Monotonicity under the decaying correlation

$$\frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}}\, t^2_{1/2}} \sum_{t\,\text{off}} t_{sd,t}\, 2^{-t_{sd,t}/t_{1/2}} > 0$$

on any path with at least one off-drug visit after an exposure.

- **Proposition 3.** On such a path, by Proposition 2 the slope's magnitude is strictly decreasing in $t_{1/2}$. By A.1, $\sigma^2_\Delta$ is strictly increasing, since $\mathrm{Var}(\Delta)$ is fixed. So the noncentrality and the power of the path-stratified statistic are strictly decreasing. On a path whose off-drug visits all precede first exposure (the placebo-first crossover path), $\bar{g}_{\text{off}} = 0$ at every half-life and the power is constant. The pooled statistic's residual variance also contains the between-path variation of mean contrasts, which depends on the half-life through the carryover means. Its monotonicity was checked on a grid (Stage 2), not proved.
- **Small half-lives.** $\bar{g}_{\text{off}} \le 2^{-1/t_{1/2}}$, which gives the behavior near zero stated in the Results.
- **Large half-lives.** $\bar{g}_{\text{off}} \to 1$ as $t_{1/2} \to \infty$, which recovers the original rule.

## A.4 The feasibility bound and the shared biomarker effect

**The feasibility bound.** Order the variables as biomarker, then responses, and partition the correlation matrix as $\begin{pmatrix} 1 & c_{bm}\tilde{g}^\top \\ c_{bm}\tilde{g} & M \end{pmatrix}$. By the Schur complement, the matrix is positive definite if and only if $M$ is positive definite and $1 - c^2_{bm}\tilde{g}^\top M^{-1}\tilde{g} > 0$. Hence the bound is $c^*_{bm} = (\tilde{g}^\top M^{-1}\tilde{g})^{-1/2}$.

**With prognostic correlations.** Let $M_0$ be the correlation matrix of everything but the biomarker, including baseline severity, which is uncorrelated with the response components. The biomarker

row is a sum of fixed patterns scaled by the correlations, $r = c_{bm}\tilde{g} + c_{PB}\,p + c_{TV}\,q + c_{BL}\,b$, where $p$ and $q$ are one on every visit of the placebo-response and natural-history components and $b$ is one on baseline severity. The same Schur argument gives the condition $r^\top M_0^{-1} r < 1$, that is $c^\top A c < 1$ with $c = (c_{bm}, c_{PB}, c_{TV}, c_{BL})$ and $A_{jk} = x_j^\top M_0^{-1} x_k$ for $x \in \{\tilde{g}, p, q, b\}$. The feasible region is the interior of an ellipsoid. The bound for a single correlation is the case $A_{\tilde{g}\tilde{g}}^{-1/2}$. Fixing all correlations but one makes $c^\top A c = 1$ a quadratic in that correlation, whose positive root is its bound. A negative discriminant means the fixed correlations are already infeasible. Because $b$ is an eigenvector of $M_0$ with unit eigenvalue, $c_{BL}$ enters only through its square. Its bound given $c_{bm}$ is $\sqrt{1 - c_{bm}^2 A_{\tilde{g}\tilde{g}}}$. The $p$ and $q$ patterns are structurally identical, so $c_{PB}$ and $c_{TV}$ have the same bounds. A design's bound is the minimum over its paths. The off-diagonal entries of $A$ are negative at the extended DGP's values, so the correlations are mildly complementary. A biomarker correlated with the drug response already implies a small correlation with the other components. An explicit correlation of the same sign therefore uses less of the feasible region than it would if the components were independent.

**The shared biomarker effect.** Let $s_0$, $s_{\text{off}}$ and $s_{\text{on}}$ be the true slopes of $Y$ on $B$ at baseline, at post-baseline off-drug visits and at on-drug visits. A model with baseline as an off-drug row estimates one $\beta_B$ for $s_0$ and $s_{\text{off}}$.

The following account is heuristic, not a derivation. Consider a prognostic biomarker with no interaction, whose post-baseline slope is constant: $s_0 = 0 \neq s_{\text{off}} = s_{\text{on}} = s_{\text{prog}}$.

- **The bias.** The fitted $\hat{\beta}_B$ lies between the baseline and off-drug slopes, $\hat{\beta}_B \approx f\, s_{\text{prog}}$ for some $0 < f < 1$ that depends on the weight the baseline row receives. The interaction absorbs the remainder, $\hat{\beta}_{bm:D} \approx (1 - f)\, s_{\text{prog}} \neq 0$.
- **The correction.** Adding $B_i \cdot \mathbf{1}\{t = 0\}$ frees $s_0$ and removes this source of bias. That the baseline row imposes such a constancy constraint is known from the literature on baseline as a covariate or a response (35).

Two further mechanisms operate in the stress-test scenarios.

- **The slope grows over time.** When the post-baseline slope grows, the bias depends on how drug status is confounded with time. Freeing $s_0$ is then not sufficient in general. The residual 0.063 for that analysis is consistent with this.

- **Expectancy.** When the biomarker is associated with the placebo response, the slope also scales with the expectancy $\eta_t$, which is higher at the open-label (on-drug) visits. The expectancy term $B_i \cdot \eta_t$, with $\eta_0 = 0$, addresses both this mechanism and the baseline constraint. Freeing $s_0$ alone does not. An expectancy-scaled slope still enters the interaction through the open-label visits.

Neither correction is exact in the Hybrid design. The expectancy term removes a constant post-baseline slope from the interaction only when the blinded on-drug and off-drug visits carry equal total weight. In the Hybrid design they do not, so a small bias remains. The corresponding derivations are in the project's compendium of mathematical results (`analysis/report/compendium-mathematics.tex`). The stress-test null rejection rates across the four prognostic cells (0.040 to 0.047 for the expectancy-term model, 0.049 to 0.063 for the free-baseline model) show that these residual biases are small at the prognostic correlations simulated.

A scenario in which the biomarker is associated with baseline severity alone was simulated in the stress test. The original analysis rejected 0.075 there, and both corrections stayed within the band.

### A.5 The separable structure

Let $A_{ts} = \rho^{|w_t - w_s|}$ and $K = (1 - c_1)I_3 + c_1 J_3$, and set $M = K \otimes A$, so that the correlation between component $c$ at visit $t$ and component $c'$ at visit $s$ is $K_{cc'}A_{ts}$.

**Validity.** The eigenvalues of $M$ are the products of those of $K$ and $A$.

- $K$ has eigenvalues $1 + 2c_1$ and $1 - c_1$ (twice), so it is positive definite for $-\frac{1}{2} < c_1 < 1$.
- $A$ is the correlation matrix of a stationary Ornstein-Uhlenbeck process at the visit times, and is positive definite for any distinct times.

Hence $M$ is positive definite for every schedule. Generatively, $\mathbf{X} = L_A \mathbf{Z} L_K^\top$ with $L_A L_A^\top = A$ and $L_K L_K^\top = K$: each column of $L_A\mathbf{Z}$ is an independent AR(1) path, and the components are fixed mixtures of them.

**The feasibility bound.** With the biomarker correlated with the drug-response component only, its correlation vector is $c_{bm}\tilde{g}$, with $\tilde{g} = e_3 \otimes g$ and $g$ the correlation profile. Since $M^{-1} = K^{-1} \otimes A^{-1}$,

$$\tilde{g}^\top M^{-1}\tilde{g} = [K^{-1}]_{33}\, g^\top A^{-1} g, \qquad [K^{-1}]_{33} = \frac{1+c_1}{(1-c_1)(1+2c_1)} = \frac{1}{1-R^2}.$$

**The schedule term.** The sampled process is Markov, with step correlation $\rho_t = \rho^{w_t - w_{t-1}}$, so

$$g^\top A^{-1} g = g_1^2 + \sum_{t=2}^{n} \frac{(g_t - \rho_t g_{t-1})^2}{1-\rho_t^2}.$$

For a binary correlation profile, each step from visit $t-1$ to visit $t$ contributes the following term:

| Step | Term |
|---|---|
| on to on | $(1-\rho_t)/(1+\rho_t)$ |
| on to off | $\rho_t^2/(1-\rho_t^2)$ |
| off to on | $1/(1-\rho_t^2)$ |
| off to off | $0$ |

Under the decaying correlation, the off-drug correlation decays instead of switching, which lowers the on-to-off term. This is why the feasibility bound under the decaying correlation rises with the half-life. Under compound symmetry it rises from 0.256 without carryover to 0.307 at 1 week and 0.369 at 2 weeks, which makes $c_{bm} = 0.3$ feasible from 1 week (Figure 5).

**The two-constant form.** With a common $\rho$, setting the same-visit cross-component correlation to $c_1$ and the different-visit correlation to $c_\times A_{ts}$ gives

$$M = K_a \otimes A + K_b \otimes I_n, \qquad K_a = (1-c_\times)I_3 + c_\times J_3, \qquad K_b = (c_1 - c_\times)(J_3 - I_3).$$

$K_b$ has zero diagonal and so is indefinite. The eigenvalues of $M$ are $\omega_j(A)(1+2c_\times) + 2(c_1 - c_\times)$ and $\omega_j(A)(1-c_\times) - (c_1 - c_\times)$, where $\omega_j(A)$ are the eigenvalues of $A$. Hence $M$ is positive definite if and only if $\omega_{\min}(A) > (c_1 - c_\times)/(1-c_\times)$. For visits at a common spacing of $s$ weeks, $\omega_{\min}(A)$ decreases toward $(1-\rho^s)/(1+\rho^s)$ as visits accumulate, which is 0.111 at weekly spacing and $\rho = 0.8$ (the threshold itself) and 0.016 at daily spacing. The values for the paper's schedules are in Stage 5.

**The original structure.** The original compound-symmetry form is $M = K_o \otimes I_n + K_p \otimes J_n$, with a person-level factor $K_p = \rho_{cs} I_3 + c_\times(J_3 - I_3)$ and an occasion-level factor $K_o = (1-\rho_{cs})I_3 + (c_1 - c_\times)(J_3 - I_3)$. It is positive definite if and only if $(1-\rho_{cs}) - (c_1 - c_\times) > 0$, whatever the schedule

(0.10 at the original values, which is also the smallest eigenvalue of the original eight-visit response block).

**The summed response.** With component standard deviations $s_t$ and $S_t = \sum_c s_{c,t} X_{c,t}$, separability gives

$$\mathrm{Cov}(S_t, S_s) = (s_t^\top K s_s)\, A_{ts},$$

proportional to the AR(1) kernel when the standard deviations are constant. The two-constant form adds $(s_t^\top K_b s_t)\, \mathbf{1}[t = s] > 0$, a white-noise term.

These results are derived in full, and checked numerically, in the pmsimstats-ng documentation (`docs/35-separable-response-covariance.md`).

### A.6 The responder-class reading of $c_{bm}$

Let $Z_i \in \{0, 1\}$ be a latent responder class with $\Pr(Z_i = 1) = p$, and let $B_i = \mu_B(Z_i) + e_{B,i}$ and $BR_i = \mu_{BR}(Z_i) + e_{BR,i}$ for the on-drug drug response, with $e_B$ and $e_{BR}$ independent of each other and of $Z$. Then $\mathrm{Cov}(B, BR) = \mathrm{Cov}(\mu_B(Z), \mu_{BR}(Z))$. Both $\mu_B(Z)$ and $\mu_{BR}(Z)$ are affine in the binary $Z$, so their correlation is $\pm 1$ and $|\mathrm{Cov}(B, BR)| = \mathrm{SD}(\mu_B(Z))\,\mathrm{SD}(\mu_{BR}(Z))$. Dividing by $\sigma_B \sigma_{BR}$,

$$c_{bm}^2 = \frac{\mathrm{Var}(\mu_B(Z))}{\sigma_B^2} \cdot \frac{\mathrm{Var}(\mu_{BR}(Z))}{\sigma_{BR}^2} = f_B\, f_{BR},$$

the product of the between-class variance fractions. With more than two classes the correlation of $\mu_B(Z)$ and $\mu_{BR}(Z)$ is at most 1 in magnitude, so $c_{bm}^2 \le f_B f_{BR}$.

## Appendix B. Code changes

### B.1 The decaying correlation in the original `generateData()` (commit 58b32a9)

The original lines (`R/generateData.R`, lines 131 to 138) set the biomarker-response correlation to $c_{bm}$ wherever the drug-response mean is nonzero:

```
# correlation with biomarker
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1,'bm']<-modelparam$c.bm
    correlations['bm',n1]<-modelparam$c.bm
  }
}
```

The replacement sets the correlation in proportion to the remaining drug effect, with zero correlation before first exposure:

```
# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (graded coupling), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(d[p]$tod>0){
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
  }else{
    g<-0
  }
  correlations[n1,'bm']<-modelparam$c.bm*g
  correlations['bm',n1]<-modelparam$c.bm*g
}
```

## B.2 Stop instead of repairing

The original code replaces a non-positive-definite matrix with a nearby valid one when its `makePositiveDefinite` option is on, as it is in the original driver (lines 145 to 150). The replacement stops, so that an infeasible cell is reported rather than changed:

```
# published:
# if(makePositiveDefinite){
#   if(!is.positive.definite(sigma)){
#     sigma<-make.positive.definite(sigma, tol=1e-3)
#   }
# }
if(!corpcor::is.positive.definite(sigma)){
  stop(sprintf('c.bm = %.3f exceeds the feasible range for this design',
               modelparam$c.bm))
}
```

One case is exempt from the stop. The placebo-response component's standard deviation is proportional to the expectancy. A visit with expectancy zero (a run-in or a baseline visit coded with no expectancy) has exactly zero placebo-response variance. It makes $\Sigma$ singular by construction rather than by an infeasible correlation. The package's stop path checks and factors only the sub-block of positive-variance components and draws the zero-variance components at their means. No correlation is modified. The original repair set those variances to $10^{-3}$ instead.

### B.3 Carryover decaying from the last on-drug mean

The original carryover rule (lines 83 to 92) decays the previous visit's mean by the total time since discontinuation. At the second and later off-drug visits that mean has already been decayed, so the decay compounds. The replacement decays from the last on-drug value. Study 1 uses it at every positive half-life. Replacing it with the original rule changes no slope, variance or feasibility bound, only the path means. It therefore changes the study 1 closed-form power by at most 0.002 in the Hybrid design. In the original pipeline the change alone moves power by at most 0.006 (`docs/53`, Section 8.3). Combined with the restart of the Gompertz clock at re-exposure, it can produce a downward step in the mean at the first re-exposure visit when the residual exceeds the restarted curve's value. On the Hybrid schedule it does not (Stage 3).

```
brmeans<-modgompertz(d$tod,rp$max,rp$disp,rp$rate)
if(nP>1){
  last_on<-0
```

```
    for(p in 1:nP){
      if(d[p]$onDrug){
        last_on<-brmeans[p]
      }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
        brmeans[p]<-last_on*(1/2)^(d$tsd[p]/modelparam$carryover_t1half)
      }
    }
  }
```

## B.4 The exposure-decayed drug indicator in the analysis

Analyses that use an exposure-decayed drug indicator must code off-drug visits before first exposure as unexposed. The time since discontinuation is zero there, and the decay function returns 1 at zero, so without a guard a placebo-first crossover participant would be coded fully exposed at every visit. In the pmsimstats-ng carryover manuscript this guard was missing. In a paired rerun of that manuscript's crossover cells (`analysis/scripts/carryover-sensitivity/26-co-dbc-fix-paired.R`, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week, 500 replicates), restoring the guard raised the power of the exposure-weighted analysis from 0.512 to 0.814, the level of the binary indicator (0.810):

```
exposure_weight <- function(Db, tsd, halflife) {
  w <- 0.5^(tsd / halflife)
  ifelse(Db == 1, 1, ifelse(tsd > 0, w, 0))
}
```

## B.5 The extended DGP in pmsimstats-ng

The package function `buildSigma()` (`R/generateData.R`) implements the decaying correlation, with `c.bm` on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure and zero before exposure. It implements AR(1) within each component in calendar weeks. It decays carryover on the drug-response mean from the last on-drug value (not the original rule of B.3). It implements the prognostic correlations of Stage 6: with the placebo-response component (`c.bm.pb`), with the natural-history component as a constant correlation (`c.bm.tv`) or as a mean shift growing along the natural-history curve (`b.bm.tv`), and with baseline severity (`c.bm.bl`). The first three enter

the biomarker row of the correlation matrix and share the joint bound of Table 9. The mean shift does not enter the matrix and has no bound.

**The cross-component block.** `buildSigma()` builds the cross-component correlation from two constants: `c.cf1t` at the same visit and `c.cfct * rho^tg` at a lag of `tg` weeks. With `c.cf1t = c.cfct = c_1` this is exactly the separable form of Table 11, $c_1\rho^{|w_t - w_s|}$ at every lag, since $\rho^0 = 1$. With the two constants unequal it is the two-constant form of Stage 5. The extended DGP is therefore the package with both constants set to 0.2, which is how the project's carryover driver calls it (`analysis/scripts/carryover-sensitivity/simulation-core.R`). Study 3's driver builds the separable block directly (`C1 * A` in `11-strawman-stress-test.R`).

**Allocation and carryover in study 3.** The driver takes `--allocation fixed|random` and `--carryover anchored|recursive`. The tables in Stage 6 are from the first run, with fixed allocation (18, 18, 17, 17) and the original carryover rule (`recursive`). The rerun used carryover decaying from the last on-drug mean (`anchored`) and randomized allocation. It changed no null rejection rate beyond Monte Carlo error and no prognostic verdict. Its power values pooled with the first run's are quoted once, in the paragraph on power and robustness. The extended DGP as defined uses fixed allocation. The rerun's randomized allocation is a driver option that is not part of the definition.

**Stop instead of repair.** `buildSigma()`, `generateData()` and `generateSimulatedResults()` accept a `stopOnNonPD` argument, `TRUE` by default. A non-positive-definite covariance matrix then raises an error, exactly as in B.2, both at the explicit positive-definiteness check and in the Cholesky-factorization fallback that previously repaired regardless of the `makePositiveDefinite` setting. The original DGP's repair is reproduced only by passing `stopOnNonPD = FALSE` and setting `options(pmsimstats.allow_repair = TRUE)` explicitly, as the scripts that study the repair itself do. Without the option, the repair is locked off: a draw from a matrix that is not positive definite stops with an error whatever the other arguments are. The argument was first added with a `FALSE` default for backward compatibility. The default was changed after two runs of the companion carryover study repaired infeasible cells and reported the attenuated interaction as a power loss.

**Not in the package.** The measurement-error, Weibull-decay and dropout options of Table 11 are implemented in the stress-test driver (`11-strawman-stress-test.R`) and in the companion study's driver core, and not yet in the package. The package's dropout mechanisms (`censordata.R`) are the original censoring patterns, with one correction. In the original code the biased term weights

each participant by 100 plus the improvement from baseline, raised to the power `eb1`. Dropout therefore rose with improvement, the reverse of the mechanism the original study describes. The package now weights by 100 minus the improvement, floored at zero, so that dropout rises with worsening. Figure 4 uses the corrected rule. Figure 3 and `implementations/original` keep the published rule.